

Identifying Treatment Effects on Productivity: Theory with an Application to Production Digitalization*

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Abstract

We study the identification and estimation of treatment effects on the productivity of firms. Our approach embeds standard methods of production function estimation into a dynamic potential outcome framework. This new framework clarifies the necessary assumptions and potential pitfalls when quantifying causal effects on productivity. Our method can be applied under weaker assumptions than those that have been previously employed in the literature and does not require solving the firm's dynamic optimization problem. We apply our method to study the effect of production digitalization on productivity growth. Our results robustly show that, except for the electronic manufacturing industry, the average treatment effect of production digitalization is not significant in a window of four years after production digitalization. However, we find substantial heterogeneity in the impact of production digitalization on productivity across time and industries. Importantly, firms with medium productivity before production digitalization tend to experience the most productivity increase over time.

Keywords: dynamic treatment effects; productivity; potential outcomes

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1 Introduction

Researchers have long been interested in quantifying the effect of an investment or intervention on a firm’s productivity.¹ A natural two-step approach would be to first estimate the firm’s productivity and then compare this with an estimate of what its productivity would have been in the counterfactual world absent the change. Problematically, however, issues can arise if one simply borrows one of the typical methods of estimating production functions and feeds the estimated productivities into a standard policy evaluation method for estimating treatment effects. In general, the issue is that both of these procedures rely on distinct sets of assumptions that may be incompatible with each other, leading to incorrect inferences about the causal effects (De Loecker and Syverson, 2021). In this article, we propose a method of estimating causal effects on productivity that fits the general two-step description, but we adapt existing methods to ensure that the assumptions invoked to estimate the realized and counterfactual productivities are consistent and sufficient to identify the treatment effect.

Specifically, when estimating productivities from firm- or plant-level data, researchers usually assume that productivity follows a Markov process (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015; Gandhi et al., 2020). Meanwhile, when estimating treatment effects on productivity, the firm’s productivity in the treated and untreated states would usually be modeled as potential outcomes. If the firm’s potential productivities in the treated and untreated states follow Markov processes, then the realized productivity may not be Markovian. For example, if the intervention of interest is the adoption of a new production technology, the plant’s potential productivities with and without the new technology might be affected by different types of unobservable inputs.² In this case, researchers may prefer to model the treated and untreated potential productivities as separate Markov processes. Accordingly, in the first period in which the firm adopts the technology, the firm’s realized productivity will be its treated productivity, whose distribution depends on the previous treated potential productivity as opposed to the previous untreated productivity that was realized in the data. As a result, the estimated productivity will be biased if the researcher employs standard methods

¹See the excellent literature reviews by Bartelsman and Doms (2000) and Syverson (2011). Empirical studies come from a wide range of fields including trade and development (e.g., Pavcnik, 2003; De Loecker, 2007; Amiti and Konings, 2007; De Loecker, 2013; Yu, 2015; Brandt et al., 2017), industrial organization (e.g., Doraszelski and Jaumandreu, 2013; Braguinsky et al., 2015), political economy (e.g., He et al., 2020; Chen et al., 2021), and public economics (Liu and Mao, 2019).

²In the application in Section 6, we consider the production digitalization of manufacturing firms, in which case a firm’s unobservable ability to deploy digitalization technologies is likely relevant to the firm’s production only after the adoption and may not be perfectly proxied by the firm’s total factor productivity before the adoption.

that assume the sequence of realized productivities is Markovian. We show that one can simply restrict attention to periods in which the plant remained treated or untreated in sequential periods in order to estimate the production function and the realized productivity in each period. In fact, one can separately estimate production functions in the treated and untreated states in order to allow for different output elasticities for inputs after the intervention.

Given consistent estimates of the plant’s realized productivity in each period under observation, the researcher’s remaining task is to estimate the “missing counterfactual,” i.e., the potential productivity that was not realized in the data. If the intervention is purely exogenous, then a standard difference-in-differences approach can be used to estimate the average treatment effect. However, the decision to export or adopt a new technology is likely to be endogenous, and the standard parallel trends assumption is likely to fail. We show that the structural assumptions invoked to estimate the productivities can be redeployed to easily solve this issue of selection into treatment. Namely, the fact that untreated productivity follows a Markov process implies that an untreated firm or plant can be matched to a treated one with the same realized productivity in the period before it was treated in order to fill in the missing counterfactual.

In comparison with earlier work, our approach is both more generally applicable and more narrowly focused on estimating the treatment effect. It is more general in the sense that earlier work estimated returns to research and development or to exporting by assuming realized productivity follows a controlled Markov process (e.g., [De Loecker, 2013](#); [Doraszelski and Jaumandreu, 2013](#); [Chen et al., 2021](#)). This assumption is more restrictive than the assumption we adopt and may not be satisfied if, for example, potential productivities follow independent Markov processes. In [Section 5](#), we discuss the potential pitfalls of using a controlled Markov process for realized productivity. At the same time, our work is more narrowly focused because we do not attempt to estimate and identify all features of the model. We are only interested in estimating a binary treatment effect, for example, the average treatment effect on the treated where the treatment would be defined as actively investing in R&D or exporting.³

Because our goals and requirements are more focused, we opt not to solve the firm’s dynamic optimization problem, which is a characteristic of the approaches in the existing empirical productivity literature ([Olley and Pakes, 1996](#); [Levinsohn and Petrin, 2003](#); [Aw et al., 2011](#); [Doraszelski and Jaumandreu, 2013](#)). To identify an average treatment effect on

³The methodology in this article does not extend to the case of continuous treatment variables. Accordingly, we would not use this framework to estimate the marginal effect of R&D expenditures or export sales. We leave this extension for future work.

the treated, we only need to identify the mean productivity in the next period conditional on the current productivity in the event that the treatment status does not change. Our approach is more typical of the literature on dynamic potential outcomes, in which the treatment selection rule is left unspecified except for some timing assumptions on firms' choices. Namely, we assume that the firm chooses its treatment status for the current period before the realization of its productivity innovation, but we allow the firm to select its treatment status based on all previous treated and untreated potential productivities. Thus, the timing of a firm's choice of treatment status is modeled in the same way that capital is in most of the production function estimation literature.

Instead of precisely modeling how firms select into treatment, we make a high-level assumption about the fraction of observations in the treated and untreated states in order to consistently estimate the production function. Specifically, we require that the data include positive fractions of observations in which firms remain untreated in consecutive periods, are untreated then become treated, and remain treated in consecutive periods. In an ideal setting for policy evaluation, we would observe two periods before and after treatment were randomly assigned among a group of firms. More generally, however, our framework allows treatment status to be endogenously determined outside of the model. For example, firms might select into treatment on the basis of an unmodeled idiosyncratic switching cost. Moreover, firms do not need to choose treatment optimally.

Our work also relates to earlier work that uses regression-based methods to estimate causal effects on productivity (e.g., [Pavcnik, 2003](#); [Amiti and Konings, 2007](#); [Yu, 2015](#); [He et al., 2020](#)). In this approach, the firms' productivities are estimated in a first step that ignores the variation in the treatment status. In the second step, regression methods are used to estimate the average effect of a policy on productivity. Because this procedure generally yields inconsistent estimators of an average treatment effect, even under the strong assumptions that realized productivity follows a Markov process and treatment is exogenously assigned, our more robust methodology provides a useful alternative. We discuss these issues in [Section 3.3](#).

Finally, [De Loecker \(2007\)](#) and [De Loecker \(2013\)](#) represent early examples of the hybrid approach we pursue in this article, which combines traditional production function estimation with policy evaluation tools. [De Loecker \(2007\)](#) estimates a production function under the assumption that productivity follows a controlled Markov process and subsequently estimates a treatment effect using one of the three methods: propensity score matching, the estimated productivity transition function, or a difference-in-differences analysis. Compared to this work, we relax the assumptions on the productivity process and emphasize the conditions under which the estimand can be interpreted as

an average treatment effect or an average treatment effect on the treated.

The rest of the article is organized as follows. In Section 2, we formally introduce a model of a firm that uses capital, labor, and intermediate inputs to produce with a Hicks-neutral productivity factor. The firm’s realized productivity is equal to one of the treated or untreated potential productivities depending on the firm’s treatment status. The potential productivities are assumed to follow a Markov process that generalizes the assumptions typically used in the literature. The key restriction is that the counterfactual productivity in the previous period is independent of the current productivity if the treatment status does not change, but we do not restrict the evolution of the potential productivities in the period in which the treatment status changes. This allows us to accommodate a wide range of plausible scenarios. As previously mentioned, the potential productivities might evolve independently of one another. Alternatively, the potential productivities might follow parallel paths, or the treated productivity path might branch from the untreated productivity path in the period in which the firm takes treatment. The researcher does not need to take a stand about the nature of the treatment in order to estimate the treatment effect.

In Section 3, we first review the identification of the production function using [Gandhi et al. \(2020\)](#) and [Akerberg et al. \(2015\)](#) in the setting in which treatment status does not change. We then show how the moment conditions must be modified to allow for variations in treatment statuses. Here, the key assumption that enables the proposed method is that treatments are selected prior to the realization of the productivity innovation. Otherwise, the assumptions and data requirements are analogous to those of [Gandhi et al. \(2020\)](#) and [Akerberg et al. \(2015\)](#): we require panel data with at least two periods and many firms. In addition, we must observe some firms that remain untreated for two consecutive periods, and another group of firms that switch from the untreated to the treated state, and remain treated for two consecutive periods. This data feature allows us to identify the production function and treatment effect. In Appendix D.1, we discuss an alternative assumption and moment conditions that might be used if this assumption on firm types does not appear to be satisfied in the data.

In Section 4, we discuss the identification of the average treatment effect on the treated (ATT). The average treatment effect (ATE) is generally not identified without stronger structural assumptions, and we thus discuss its identification in Appendix A.2. In contrast to the literature on dynamic treatment effects ([Heckman and Navarro, 2007](#); [Abbring and Heckman, 2007](#); [Vikström et al., 2018](#); [Sun and Abraham, 2021](#)), the outcome of interest is not directly observed, and additional structural assumptions are needed to infer it from data. Apart from this distinction, our approach builds on the dynamic potential

outcome framework. As has been observed in the literature, one must be careful when defining and identifying treatment effects in a dynamic potential outcome framework because firms who are treated in one period may return to the untreated state but continue on an altered trajectory as a result of their temporary treatment assignment. We do not add to this discussion, but acknowledge the complexities involved. For the sake of simplicity, we focus on the case of an absorbing treatment state in which firms remain forever after they first select into treatment. Accordingly, our identification and estimation results target the ℓ -period ahead average treatment effect on the treated, which answers the question of how much more or less productive a firm is ℓ periods after it is initially treated compared to what its productivity would have been if it had remained untreated the entire time.

In Section 5, we discuss the ramifications of placing the Markov assumption on the potential productivities as opposed to the realized productivities. We show that the controlled Markov process assumption on the realized productivity omits important unobserved heterogeneity, and it will lead to a misleading causal interpretation of the transition period dynamics and hence a misleading ATE.

In Section 6, we use our methodology to estimate the productivity effects of production digitalization in the manufacturing sector of China. Recent work has struggled to find evidence in aggregate production statistics of any productivity gains associated with the rise of AI-related production technologies (Brynjolfsson et al., 2017). In contrast with our approach, existing research relies on reduced-form regressions on the productivity estimates to detect the impact of the adoption of new technologies (Draca et al., 2009; Gal et al., 2019, among others). Our empirical analysis of firm-level data shows that, in a window of four years, the average productivity gains from production digitalization are positive in the first two periods after production digitalization but are negative in later periods. The overall productivity effects are mildly negative and not statistically significant. Moreover, the productivity effects of digitalization vary substantially across industries. The greatest productivity gains are realized in industries that use more complex production processes like electronics, equipment, and healthcare. In particular, we find a positive and significant ATT on productivity for the electronics manufacturing industry. The results on the firm-specific productivity effects reveal that the effects of digitalization on productivity become more dispersed as time evolves. Importantly, we also find that firms with a medium level of pre-treatment productivity receive the largest productivity gains over time. In general, our results support the view that new digital technologies have unequal effects on firms' productivities depending on the firm's characteristics (Bloom et al., 2012; Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Using the same dataset, we also show that regression-based methods may lead to qualitatively different conclusions. In fact, focusing on the industry of electronics manufacturing, when we pattern our analysis off of existing methods, depending on the productivity process specification, we find quantitatively different results. This conflicting result is not surprising in light of the specification issues with regression-based approaches and the substantial heterogeneity revealed by our analysis, but it demonstrates the potential for quantitatively significant discrepancies between existing approaches and ours.

Finally, Section 7 concludes.

All proofs are collected in Appendix B.

2 The Econometric Framework

2.1 A Firm Model with Treatment and Potential Productivity

Firms produce with a Hicks-neutral production technology. The evolution of firms' productivity can be affected by a binary treatment. Let D_{it} denote the treatment indicator $D_{it} \in \{0, 1\}$, with $D_{it} = 1$ indicating that firm i receives the treatment in period t . The treatment can be imposed externally (e.g., trade liberalization, environmental regulations, etc.) or chosen by the firm (e.g., digitalization, R&D investment, importing, exporting, etc.). In period t , firm i has the following production function

$$Q_{it} = e^{\omega_{it} + \eta_{it}} F(K_{it}, L_{it}, M_{it}; \beta), \quad (1)$$

where Q_{it} denotes the output quantity, ω_{it} denotes the realized productivity, η_{it} denotes an ex-post innovation to productivity that is not known when a firm chooses inputs for time t , K_{it} denotes capital inputs, L_{it} denotes labor inputs, M_{it} denotes material inputs, and β is a parameter vector. The dimension of β can be infinite when the production function is nonparametric. Moreover, β can also include time-related variables to account for secular trends in the production function (e.g., [Doraszelski and Jaumandreu \(2013\)](#)).⁴

The two potential outcomes of productivity are ω_{it}^0 and ω_{it}^1 . The binary treatment D_{it} determines the realized productivity through the following equation:

$$\omega_{it} = \omega_{it}^1 D_{it} + \omega_{it}^0 (1 - D_{it}). \quad (2)$$

⁴In Appendix C.1, we discuss the type of treatment that can have impact on the organization of production or managerial efficiency such that the treatment affects the shape of the production function ([Chen et al., 2021](#)). We introduce two approaches to adjust for the differences in the pre- and post-treatment production functions, one uses the indexing number method ([Caves et al., 1982](#); [Brandt et al., 2012](#)), and the other uses the industrial average inputs to benchmark the production functions.

The firm may know its potential productivities when making decisions, but the econometrician does not directly observe either potential outcome. Accordingly, our model admits the possibility that firms select into treatment on the basis of the unobservable productivity (both ω_{it-1}^1 and ω_{it-1}^0), as opposed to selection only on the realized productivity ω_{it-1} . For simplicity, we focus on the absorbing treatment case, i.e., $D_{it} \geq D_{it-1}$.

Conventionally, the realized productivity is assumed to follow a first-order Markov process (Olley and Pakes, 1996; Levinsohn and Petrin, 2003; Akerberg et al., 2015), but we generalize this assumption with the following Markov process for $(\omega_{it}^1, \omega_{it}^0)$:

$$\begin{aligned}\omega_{it}^1 &= \mathbb{1}(D_{it} = D_{it-1})\bar{h}_1(\omega_{it-1}^0, \omega_{it-1}^1) + \mathbb{1}(D_{it} > D_{it-1})h_1^+(\omega_{it-1}^0, \omega_{it-1}^1) + \epsilon_{it}^1, \\ \omega_{it}^0 &= \mathbb{1}(D_{it} = D_{it-1})\bar{h}_0(\omega_{it-1}^0, \omega_{it-1}^1) + \mathbb{1}(D_{it} > D_{it-1})h_0^+(\omega_{it-1}^0, \omega_{it-1}^1) + \epsilon_{it}^0,\end{aligned}\tag{3}$$

where we distinguish the evolution process during the first treatment period $D_{it} > D_{it-1}$ and the non-transition period $D_{it} = D_{it-1}$ period. In (3), h_1^+ and h_0^+ are the transition functions of ω_{it}^1 and ω_{it}^0 when the firm becomes newly treated, and $\bar{h}_1$ and $\bar{h}_0$ are the transition functions when the treatment status is unchanged.⁵

We further impose the following assumption to make sure that our potential productivity process (3) is consistent with the specification in the existing literature when no variation in treatment is assumed:

Assumption 2.1. (*Diagonal Markov Process*) The function $\bar{h}_d$ depends only on ω_{it}^d , so we may abuse notation to rewrite

$$\bar{h}_d(\omega_{it}^0, \omega_{it}^1) = \bar{h}_d(\omega_{it}^d),$$

and $\mathbb{E}[\epsilon_{it}^d | \omega_{it-1}^0, \omega_{it-1}^1] = 0$ for $d = 0, 1$.

Assumption 2.1 says that the evolution of the potential outcome ω_{it}^0 does not depend on ω_{it}^1 if the firm's treatment status does not change. This assumption generalizes the productivity processes previously considered in the literature because, when $D_{it} = D_{it-1}$ for all i and t , each firm's productivity satisfies $\omega_{it}^d = \bar{h}_d(\omega_{it-1}) + \epsilon_{it}^d$, for $d \in \{0, 1\}$. Therefore, we can think of the conventional Markov productivity process as the special case in which there is no treatment, i.e., $D_{it} = 0$ for all i and t .

Importantly, the process (3) allows for non-Markovian behavior in the realized productivity ω_{it} . In particular, during the treatment-switching period, the distribution of the realized (treated) productivity ω_{it} may depend on both ω_{it-1}^1 and ω_{it-1}^0 . Otherwise, the distribution of the realized productivity depends on the lagged realized productivity

⁵We will introduce another transition function h^- to govern the transition out of treatment when we later consider the case of a non-absorbing treatment. See Appendix C.3 for more details.

through distinct transition functions $\bar{h}_0$ and $\bar{h}_1$ in the untreated and treated states. As a result, the distribution of productivity depends on more than the lagged realized productivity. The latter source of non-Markovian behavior could be resolved by adding D_{it} as an additional state variable in the Markov process. However, this does not eliminate the non-Markovian behavior due to the possibility that ω_{it} depends on both potential productivities in the transition period.

A wide range of models can be viewed as special cases of the generalized productivity evolution process (3). Below, we list three examples of productivity processes that satisfy equation (3), though the econometrician does not need to assume that the evolution process fits any one of these narratives. The truth can be something of a mixture of the following or other more exotic productivity processes.

Example 1. (*Parallel-Shifted Productivity Processes*) A policy might simply shift the productivity process by a constant amount. This can be accommodated by assuming potential productivities almost surely satisfy $\omega_{it}^1 - \omega_{it}^0 = C$ for some constant C . More precisely, we would assume that potential productivities (i) have an initial difference of C , i.e., $\omega_{i1}^1 = \omega_{i1}^0 + C$ almost surely for some constant C , (ii) are affected by the same innovations $\epsilon_{it}^1 = \epsilon_{it}^0$ almost surely for all t , and (iii) evolve according to transition functions that satisfy $\bar{h}_1 = h_1^+$, and $\bar{h}_1(\omega) = \bar{h}_0(\omega - C) + C$.

Example 2. (*Diverging Productivity Processes*) Consider a case where the binary treatment represents whether a firm invests in R&D. If a firm first begins investing in R&D at $t+1$, then we might assume that only ω_{it}^0 affects ω_{it+1}^1 . This assumption can be modeled by $h_1^+(\omega_{it}^0, \omega_{it}^1) = h_1^+(\omega_{it}^0)$. Because for any period before initial treatment $s \leq t$, the treated potential productivity ω_{is}^1 is irrelevant to the firm's future outcomes, we can, without loss of generality, impose $\omega_{is}^1 \equiv \omega_{is}^0$ for all pre-treatment periods $s \leq t$.

Example 3. (*Independent Productivity Evolution*) In some cases, a firm chooses between two types of production technologies. We might assume that the firm's total factor productivity using either technology evolves without being influenced by the other and that the firm simply jumps from one productivity path to the other when it switches the technology. We would impose the additional restriction that $\bar{h}_1 = h_1^+$ and $\bar{h}_0 = h_0^+$.

We note that the second example could be equivalently modeled in terms of realized productivities as in $\omega_{it} = \tilde{h}(\omega_{it-1}, D_{it}, D_{it-1}) + \epsilon_{it}$ without referring to the counterfactual potential productivities.⁶ In Section 5, we formally compare the Markov process for potential productivity in equation (3) to this alternative: We observe that $\omega_{it} =$

⁶The only difference with earlier work that treats realized productivity as a controlled Markov process would then be that we include a lag of the treatment status indicator in the transition function to allow initial treatment effects and subsequent effects to be different.

$\tilde{h}(\omega_{it-1}, D_{it}, D_{it-1}) + \epsilon_{it}$ imposes an additional structural assumption that rules out an important source of unobserved heterogeneity of treatment choice decision and it may lead to a misleading interpretation of treatment effects on productivity.

Finally, we observe that alternative generalizations of (3) would be difficult to accommodate without strengthening the assumptions commonly used to estimate production functions. For example, one might wish to relax Assumption 2.1 and allow the potential productivity ω_{it}^1 to depend on both the potential productivities in all periods. In general, however, we would not be able to identify the production function from the observed inputs and outputs because the evolution of productivity would be influenced by the unobserved counterfactual productivity. Unlike the exogenous productivity innovations, this latent productivity would be correlated with the firm's input choices if the firm anticipates that its treatment status might change in the future, which would invalidate the moment conditions we use to identify the production function. Thus, in addition to highlighting the important distinction between the evolution of realized productivity and potential productivities, the dynamic potential outcome framework in this article elucidates the assumptions required to identify causal effects on productivity.

Firms' Behavior and Timing of Firms' Decisions

Following Akerberg et al. (2015) and Gandhi et al. (2020), we distinguish the static inputs from the predetermined inputs.

Assumption 2.2. (*Timing of Inputs*) Capital K_{it} and labor L_{it} are determined at or before $t - 1$. Intermediate input M_{it} is determined no sooner than the other inputs after the realization of ω_{it} .

The treatment variable can be either determined by the external environment or chosen by the firm. We distinguish between these two cases and make the following assumption on its timing.

Assumption 2.3. (*Timing of Treatment*) (1) When the treatment is externally imposed, D_{it} is determined at or before $t - 1$; (2) When the treatment is a firm choice, D_{it} is chosen before $(\omega_{it}^0, \omega_{it}^1)$ is realized but after observing $(\omega_{it-1}^0, \omega_{it-1}^1)$.

Firms make two types of choices at the time t . First, given the realized productivity ω_{it} and predetermined inputs, firms choose the static inputs to maximize their short-run revenue. Then, firms choose the next-period predetermined inputs and possibly the treatment status D_{t+1} in period $t + 1$ given the history of state variables $S_{it} \equiv (K_{it}, L_{it}, D_{it}, \omega_{it}^1, \omega_{it}^0, \zeta_{it})$, where ζ_{it} is an idiosyncratic cost shock relevant to the dynamic decision⁷. We summarize the timeline of the firm decision in the Figure 1 when the labor

⁷For example, ζ_{it} can be idiosyncratic costs of undertaking the treatment.

is predetermined.

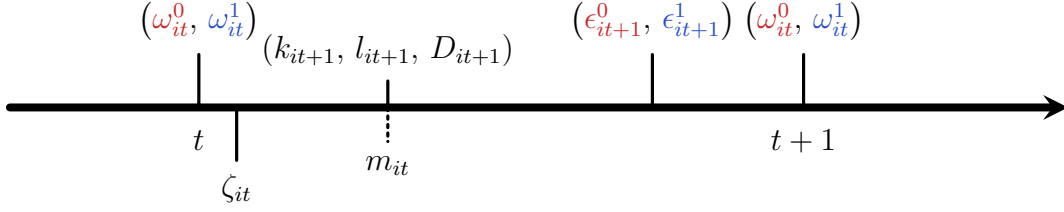


Figure 1: Timeline for a firm's decision.

2.2 Treatment Effects

Because we focus on the absorbing treatment case, the firm's treatment status can be summarized by the initial treatment time e_i , i.e., $D_{is} = 0$ for all $s < e_i$ and $D_{is} = 1$ for all $s \geq e_i$. We define the ℓ -period ahead treatment effect for firm i as $\omega_{ie_i+\ell}^1 - \omega_{ie_i+\ell}^0$ for any $\ell \geq 0$. When $\ell = 0$, the firm first switches from the untreated to the treated status, and this is the instantaneous effect coming from switching to a different process. Any subsequent dynamic treatment effect $\omega_{ie_i+\ell}^1 - \omega_{ie_i+\ell}^0$ is the composition of the instantaneous effect and the accumulation of the evolution difference between $\bar{h}_1$ and $\bar{h}_0$.

3 Recovering the Unobserved Productivity

To recover a firm's productivity, we assume that the econometrician observes a panel of inputs and outputs:

Assumption 3.1. *The econometrician has instruments $\mathcal{Z}_{it} = \{K_{it}, L_{it}, (D_{is}, K_{is-1}, L_{is-1}, M_{is-1})_{s \leq t}\}$. Moreover, $E[\epsilon_{it}|\mathcal{Z}_{it}] = 0$, and $E[\eta_{it}|\mathcal{Z}_{it}, M_{it}] = 0$.*

Assumption 3.1 is standard in the classical production function estimation literature, and is justified by the assumption on the timing of the firm's decisions.

3.1 Recovering the Productivity in the Absence of Treatment

We first review the case where $D_{it} = 0$ for all i and t , i.e., there is no treatment at all. There are two strands of literature that use different moments to identify the production functions. For the gross output production function, we follow GNR (Gandhi et al., 2020) and use a first-order condition based on the profit-maximizing choice of material inputs. For the value-added production function, we follow ACF (Akerberg et al., 2015) and

proxy for productivity using materials and other inputs. We use the lower and upper case letters to represent logs and levels of the corresponding variables, respectively. We focus on the predetermined labor in our baseline model (Assumption 2.2) but different timing assumptions on labor can be accommodated by adjusting the moment conditions.⁸

GNR Approach to Gross Production Functions When firms are output price takers, the GNR approach uses the following equation for the expenditure share of materials derived from the firm's profit-maximizing problem:

$$\mathbb{E} \left[s_{it} - \log \left(\frac{\partial f(k_{it}, l_{it}, m_{it}; \beta)}{\partial m_{it}} \right) \middle| \mathcal{Z}_{it}, m_{it} \right] = 0 \quad \forall t = 1, \dots, T, \quad (4)$$

where f is the log production function and s_{it} is the log material-cost-to-revenue-ratio. This moment condition identifies the elasticity of output with respect to materials. To recover the other parameters of the production function and the productivity evolution process, we then use Assumption 3.1 and specialize equation (2) to the case in which $D_{it} = 0$ for all i and t to obtain an additional moment condition:

$$\mathbb{E}[\omega_{it}(\beta) - h(\omega_{it-1}(\beta)) | \mathcal{Z}_{it}] = 0 \quad \forall t = 1, \dots, T, \quad (5)$$

where $\omega_{it}(\beta) = q_{it} - f(k_{it}, l_{it}, m_{it}; \beta)$ denotes the logged productivity implied by β .

ACF Approach to Value-added Production Functions Consider the logged value-added production function $f(k_{it}, l_{it}; \beta)$. The material m_{it} is a strictly monotone function of ω_{it} and hence the nonparametric inversion $\omega_{it} = g(k_{it}, l_{it}, m_{it})$ exists. [Ackerberg et al. \(2015\)](#) first identify the nonparametric object

$$\Phi_{it-1}(k_{it-1}, l_{it-1}, m_{it-1}) \equiv \mathbb{E}[q_{it-1} | k_{it-1}, l_{it-1}, m_{it-1}], \quad (6)$$

and use the moment condition

$$E[\omega_{it}(\beta) - h[\Phi_{it-1}(k_{it-1}, l_{it-1}, m_{it-1}) - f(k_{it-1}, l_{it-1}; \beta)] | \mathcal{Z}_{it}] = 0. \quad (7)$$

When treatment status does not vary, the econometrician can use either set of moment conditions to nonparametrically identify a gross or value-added production function and

⁸For example, in the GNR method, the static labor assumption requires an additional FOC moment condition for labor; in the ACF method, if labor is static, then nonparametric identification of the production function is not feasible but parametric identification can be achieved if we can find an instrument, such as wage, for the labor input.

a sequence of total factor productivities for each firm.

Lemma 3.1. *If there is no treatment in the model, then (i) the moment conditions (4) and (5) identify the gross production function β nonparametrically up to a constant difference; and (ii) the moment conditions (6) and (7) identify the value-added production function β nonparametrically up to a constant difference. Moreover, h is identified nonparametrically using either method.*

In Appendix B, we adapt the approach of partial differential equations in the [Gandhi et al. \(2020\)](#) to prove the identification of the ACF model.

3.2 Recovering the Productivity with Variation in Treatment Status

We now extend the identification result to the case with a policy intervention. Although the treatment can be chosen by the firm, we assume that treatment is exogenous with respect to the unanticipated productivity innovations. $(\epsilon_{it}^1, \epsilon_{it}^0)$.

Assumption 3.2. *(Conditional Mean-Zero Innovations) The productivity innovations $(\epsilon_{it}^0, \epsilon_{it}^1)$ satisfy*

$$\mathbb{E}[(\epsilon_{is}^0, \epsilon_{is}^1) | \mathcal{Z}_{it}] = \mathbf{0}, \quad \forall s \geq t.$$

Notably, Assumption 3.2 allows treatment assignment to depend on past potential outcomes ω_{is}^0 and ω_{is}^1 for $s < t$. To identify treatment effects, we will need to assume that there exists a positive fraction of firms that take treatment and remain treated for at least two periods.

Assumption 3.3. *There exist two periods t_0, t_1 such that $\Pr(D_{it_0} = D_{it_0-1} = 0) \neq 0$, $\Pr(D_{it_1} = D_{it_1-1} = 1) \neq 0$.*

Note that the periods t_0 and t_1 may be equal to each other.

Theorem 3.1. *Suppose Assumptions 2.1- 3.3 hold. The moment condition (4) (and respectively (6)) and*

$$\mathbb{E}[\omega_{it}(\beta) - \bar{h}_0(\omega_{it-1}(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] = 0, \quad (8)$$

$$\mathbb{E}[\omega_{it}(\beta) - \bar{h}_1(\omega_{it-1}(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 1] = 0, \quad (9)$$

identify the gross (and respectively value-added) production function parameter β and the evolution processes $\bar{h}_0$ and $\bar{h}_1$ nonparametrically up to a constant.

To implement the moment conditions in Theorem 3.1, we must exclude the transition periods from estimation. In the case of long panel datasets, this exclusion results in minimal loss of information and poses little concern. However, for short panels, transition periods may constitute a substantial portion of the available observations, potentially compromising the validity and precision of production function estimates. To address this

issue, we propose additional moment conditions that can be applied during the transition periods under specific empirical settings. These extensions are detailed in Appendix D.1.

Other Structural Objects In some applications, the researcher may be interested in other structural objects of the model, such as the transition function h_1^+ . This object is generally not identified under the assumptions in Theorem 3.1 because we cannot simultaneously observe ω_{it}^1 and ω_{it}^0 , but would be identified under stronger assumptions, e.g., the assumption in Example 2.

By recommending our approach, we do not suggest that these objects are always less important than the treatment effects we consider in this article. Researchers might also be interested in simulating a counterfactual policy intervention, which would require h_1^+ as well as a model for how firms choose their treatment status. If, however, a treatment effect of an observed intervention is one of the researcher’s primary interests, we suggest performing this component of the analysis using our methodology as a more robust alternative to existing methods.

3.3 Revisiting Existing Methods

A simple example illustrates the limitations of two common methods of recovering productivity in the context of a time-varying treatment. Suppose that a policy is implemented between measurement times T_0 and $T_0 + 1$ and exogenously affects a random subset of firms. In general, both methods yield inconsistent estimates of the productivities under the general framework of this article. In this section, we illustrate the intuition of the misspecification in the two methods and leave the rigorous proof to Appendix D.4.

The Ex-post Regression The ex-post regression method proceeds in two steps. First, it estimates the production function and productivity evolution process ignoring the existence of the policy via (4) and (5). Second, it recovers estimated productivity $\hat{\omega}_{it}$ using $\hat{\beta}$ and $\hat{h}$, and regresses the $\hat{\omega}_{it}$ on the treatment indicator with time- and group-effects to obtain the standard difference-in-differences estimate.

The ex-post regression method will be inconsistent whenever the policy affects the evolution of productivity. In the left panel of Figure 2, the true potential evolution processes are given by the black and blue lines but the ex-post regression method estimates the red-dashed line. The ex-post regression method will only be correctly specified if the ω_{it}^1 and ω_{it}^0 processes overlap, which means that treatment has no effect on productivity.

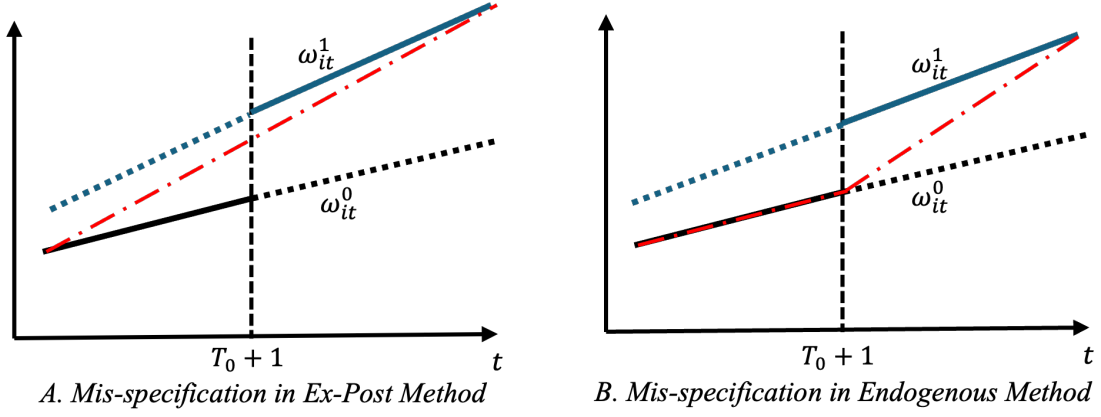


Figure 2: Misspecification in the two methods

The misspecification leads to inconsistent $\hat{\omega}_{it}$, which invalidates subsequent causal analyses.

The Endogenous Productivity Method The endogenous productivity method in [De Loecker \(2007\)](#) and [Doraszelski and Jaumandreu \(2013\)](#) includes the treatment variable in the productivity process:

$$\omega_{it} = \tilde{h}(\omega_{it-1}, D_{it}) + \epsilon_{it}.$$

After estimating the productivity process, the difference between $\tilde{h}(\cdot, 1)$ and $\tilde{h}(\cdot, 0)$ is usually regarded as the effects of D_{it} on the productivity process.

The endogenous productivity method introduces bias into the estimation of the treated productivity process $\tilde{h}(\cdot, 1)$ whenever there is a discontinuity between ω_{it}^1 and ω_{it}^0 at the time of treatment. As illustrated in the right panel of Figure 2, this method overestimates the treated evolution path by conflating the positive instantaneous treatment effect $\omega_{iT_0+1}^1 - \omega_{iT_0+1}^0$ with the dynamic process $\tilde{h}(\cdot, 1)$. In other words, it incorrectly attributes the one-time jump in productivity to a persistent shift in the evolution process. The endogenous productivity method is only correctly specified if the treated and untreated productivity paths coincide immediately before treatment, i.e., $\omega_{iT_0+1}^1 = \omega_{iT_0+1}^0$. As a result, the interpretation of treatment effects becomes ambiguous and potentially misleading. We further elaborate on this issue in Section 5. Before that, we first focus on our proposed method for identifying the treatment effect on productivity.

4 Evaluating the Treatment Effect on Productivity

For the treatment effect to be well-defined, we require a non-zero probability of treatment-switching firms. Because we only observe each firm in either the treated or untreated state, the individual treatment effect $\omega_{it}^1 - \omega_{it}^0$ is typically not identified. Consequently, we focus on the average treatment effect on the treated (ATT), which is more tractable under standard identification strategies. In contrast, identifying the average treatment effect (ATE) is generally more challenging and requires stronger structural assumptions. We discuss the identification of the ATE in Appendix A.2. The first step in our analysis is to recognize that the realized productivities are identified as a corollary to Theorem 3.1.

Corollary 4.1. *Under Assumption 2.1-3.3, we can recover the realized ex-post productivity ω_{it}^d for firms such that $D_{it} = d$.*

For ATT, we find it instructive to focus on the identification of the treatment effects under absorbing treatment, $D_{it} \geq D_{it-1}$.⁹ We leave the discussion of the non-absorbing treatment case to Appendix C.3.

Recall that $e_i > 1$ is the first period that firm i starts to receive the absorbing treatment and $D_{it} = 1$ for all $t \geq e_i$. We maintain Assumption 3.2 on the exogeneity of productivity innovations. Let $\ell \geq 0$ be the time relative to the first treatment period. The ℓ -period-ahead ATT is given by:

$$\text{ATT}_\ell = \mathbb{E}[\omega_{it}^1 - \omega_{it}^0 | t = e_i + \ell, e_i < \infty]. \quad (10)$$

The condition $e_i < \infty$ excludes the never treated firms and the ATT is conditional on the treated firms. The ATT's definition (10) does not distinguish firms that receive treatment earlier or later. The different treatment effects between early- and late-treated firms are usually referred to as the cohort effects. We leave the further discussion on cohort effects to Appendix C.2.

Our analysis of the treatment effect crucially relies on the following assumption on the productivity process (3):

Assumption 4.1. (Conditional Parallel Trend) $h_0^+(\omega_{it}^0, \omega_{it}^1) = \bar{h}_0(\omega_{it}^0)$.

⁹The absorbing treatment is at the core of the literature on estimating dynamic treatment effects (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Athey and Imbens, 2022). For any treatment that is not absorbing, we can replace the treatment status D_{it} with an indicator for ever having received the treatment to obtain a new treatment that is absorbing. For example, Deryugina (2017) defines the treatment to be "having had any hurricane" and investigates its impact on the fiscal cost for a county.

Assumption 4.1 relies on the implicit assumption that the firm's potential productivity continues to evolve as it would have if the firm had not been treated.¹⁰ This assumption does not have any empirical content when treatment is absorbing, because the untreated potential productivity is independent of the firm's decisions once it takes treatment. In the more general case where the treatment is non-absorbing, however, we must make an additional assumption. We delegate the discussion of the non-absorbing treatment case to Appendix C.3.

Failure of the Simple Parallel Trend Assumption Even when the treatment is not randomly assigned, the difference-in-differences (DID) method allows us to identify the ATT if a parallel trend assumption is satisfied. We first discuss the fallacy of the conventional DID assumption, and then provide a remedy.

Condition 1. (*Simple Parallel Trend*) The following is the simple parallel trend condition at period t :

$$\mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i = t] = \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i > t]. \quad (11)$$

If condition (11) holds, then the ATT_0 is identified as $\mathbb{E}[\omega_{it} | e_i = t] - \mathbb{E}[\omega_{it-1} | e_i = t] - (\mathbb{E}[\omega_{it} | e_i > t] - \mathbb{E}[\omega_{it-1} | e_i > t])$. This is the usual DID estimator for the two-period case. However, Condition 1 is imposed on the before-after change in the potential productivity before and after the treatment, and it is hard to justify. In particular, from the productivity process (3) and Assumptions 3.2 and 4.1, we can derive

$$\begin{aligned} \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i = t] &= \mathbb{E}[\bar{h}_0(\omega_{it-1}^0) - \omega_{it-1}^0 | e_i = t] \\ \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i > t] &= \mathbb{E}[\bar{h}_0(\omega_{it-1}^0) - \omega_{it-1}^0 | e_i > t]. \end{aligned} \quad (12)$$

From (12) we see that the parallel trend condition will fail if firms select into treatment based on its ex-ante productivity ω_{it-1}^0 .

The Conditional Parallel Trend Assumption We note that, by further conditioning on the value of ω_{it-1}^0 in equation (12), we have

$$\begin{aligned} \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i = t, \omega_{it-1}^0] &= \bar{h}_0(\omega_{it-1}^0) - \omega_{it-1}^0, \\ \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i > t, \omega_{it-1}^0] &= \bar{h}_0(\omega_{it-1}^0) - \omega_{it-1}^0. \end{aligned} \quad (13)$$

The two equations in (13) coincide as a result of Assumption 3.2. Therefore, the structural Assumption 3.2 implies the conditional parallel trend condition in the treatment effect

¹⁰This assumes, for example, that the firm does not lose any intangible capital specific to the untreated production technology.

literature: $\mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i = t, \omega_{it-1}^0] = \mathbb{E}[\omega_{it}^0 - \omega_{it-1}^0 | e_i > t, \omega_{it-1}^0]$. The conditional parallel trend assumption allows us to first identify the conditional ATT and subsequently the overall ATT.

Identifying Treatment Effects Based on the implication of (13), we study the ℓ -period-ahead conditional average treatment effect on the treated (CATT):

$$\text{CATT}_\ell(\omega) = E[\omega_{it}^1 - \omega_{it}^0 | t = e_i + \ell, \omega_{ie_i-1} = \omega],$$

which is further conditional on the pre-treatment realized productivity. By the law of iterative expectation, $\text{ATT}_\ell = E[\text{CATT}_\ell(\omega_{ie_i-1})]$, and we can identify ATT_ℓ if $\text{CATT}_\ell(\omega)$ is identified.

Proposition 4.1. *Under Assumption 4.1, the instantaneous CATT is identified as $\text{CATT}_0(\omega) = \mathbb{E}[\omega_{ie_i} - \bar{h}_0(\omega_{ie_i-1}) | \omega_{ie_i-1} = \omega]$. Consequently, the instantaneous ATT is identified as $\text{ATT}_0 = \mathbb{E}[\omega_{ie_i} - \bar{h}_0(\omega_{ie_i-1})]$.*

To identify the ℓ -period-ahead CATT and ATT for $\ell \geq 1$, we must recover the untreated potential outcome $\omega_{i,e_i+\ell-1}^0$. Following the logic of Proposition 4.1, we formalize below a set of assumptions that enable the identification of the ℓ -period-ahead ATT.

For notation purposes, let $\bar{h}_0^{(\ell)}$ be the ℓ -period productivity transition process, we can write $\omega_{ie_i+\ell}^0 = \bar{h}_0^{(\ell)}(\omega_{ie_i}^0, (\epsilon_{is}^0)_{s=e_i}^{e_i+\ell})$. We now impose a constraint on the productivity innovations but relax the shape restriction of $\bar{h}_0$.

Assumption 4.2. *There is a time s and a group of firms g such that (i) all firms in g are untreated by ℓ periods since time s , i.e., $e_i > s + \ell$, for $s \geq 2$, $\ell \geq 0$; and (ii) the conditional distribution of $(\epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0) | \omega_{ie_i-1}^0$ is the same as the conditional distribution of $(\epsilon_{is}^0, \dots, \epsilon_{is+\ell}^0) | (\omega_{is-1}^0, i \in g)$.*

For treated firms, group- g firms in period s are chosen by the econometrician to serve as the control group: If Assumption 4.2 holds, group- g firms in period s can serve as the match for the treated firms based on ω_{it-1}^0 and help identify the treatment effect of interest on productivity. To accommodate the possible nonlinearity in $\bar{h}_0^{(\ell)}$, Assumption 4.2 requires more than the mean-zero condition of the innovations.

We now present an identification result for the ℓ -period-ahead ATT in the following proposition.

Proposition 4.2. *Suppose Assumption 4.1 and 4.2 hold. Then the ℓ -period-ahead CATT is identified as $\text{CATT}_\ell(\omega) = \mathbb{E}[\omega_{ie_i+\ell} | \omega_{ie_i-1} = \omega] - \mathbb{E}[\omega_{is+\ell} | \omega_{is-1} = \omega, i \in g]$. The corresponding ATT is identified as $\text{ATT}_\ell = \mathbb{E}[\text{CATT}_\ell(\omega_{ie_i-1})]$, where the expectation is taken over the distribution of ω_{ie_i-1} .*

Proposition 4.2 essentially requires us to match the lagged productivity for treated firms with s -period firms. This is because we cannot observe the untreated productivity innovation ϵ_{it}^0 for treated firms and the higher order moments of ϵ_{it}^0 matter for the ℓ -period evolution process $\bar{h}_0^{(\ell)}$. As an example of the empirical application for which the matching period s can be determined, consider the following scenario:

Example 4. *In many empirical settings, researchers are interested in a cohort of firms that start their treatment in period $e_i = g_0$. For the ℓ -period-ahead ATT with $\ell \geq 0$, the researcher may use firms that are not treated until period $g_0 + \ell + 1$ as the control: $g = \{i' : e_{i'} > g_0 + \ell\}$ and set the time $s = g_0$. This means that we use not-yet-treated observations in the same time window as the control group. This choice reflects the belief that the productivity innovations of the not-yet-treated firms during the same period are similar to the productivity innovations of the treated firms after the treatment takes place.*

Intuitively, Assumption 4.2 holds if the following conditions are met. Before period g_0 , no firms were treated. At time g_0 , firms decide whether to take the treatment. Between g_0 and $g_0 + \ell + 1$, firms cannot change their treatment status due to governmental regulations or contracts. In this empirical setting, at the time g_0 , the firm makes the decision on whether the initial treatment time $e_i = g_0$ or $e_i > g_0 + \ell$, and firms can only make treatment choices based on their information set at period g_0 , which does not contain information on future innovations $(\epsilon_{ig_0}^0, \dots, \epsilon_{ig_0+\ell}^0)$. This example can be seen in many government policy reforms that gradually roll out in several phases (Chen et al., 2021).

However, if all firms can choose the initial treatment time freely after g_0 , then Assumption 4.2 typically fails. Firms that choose not to be treated until $g_0 + \ell + 1$ are likely those with higher potential productivity $(\epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0)$ and, therefore, are reluctant to switch to be treated. In this case, we can choose $g = \{\text{all firms}\}$ and $s = 2$ in Assumption 4.2. In other words, we use all firms and periods before the initial treatment period g_0 , and match firms based on their initial productivity ω_{i1} .

To identify ATT_ℓ from Proposition 4.2, we need to match a treated firm with an untreated firm with the same ex-ante productivity. This matching procedure can be difficult to implement due to two reasons. First, we may not be able to find a (g, s) pair that satisfies the independence restriction. Second, even if (g, s) is found, we may not have enough observations in group-time pair (g, s) . Moreover, if all firms are treated at period $g_0 + \ell$, we cannot identify the $\text{ATT}_{g_0+\ell+s}$ for all $s > 0$. In such cases, we can rely on the stronger assumption that productivity innovations are independent and identically distributed:

Assumption 4.3. (i) *The productivity innovations satisfy $\epsilon_{it}^0 \sim_{i.i.d.} G_\epsilon^0(\cdot)$, where the i.i.d is over both firm index i and time index t ; (ii) *We can find a group-time pair (g', s) such that all firms in g'**

are untreated in period s , and there is no selection on productivity innovations: $\epsilon_{i's}^0 | i' \in g' \sim G_\epsilon^0(\cdot)$ and $\epsilon_{it}^0 \sim G_\epsilon^0(\cdot)$ for all $t \geq e_i$.

This assumption allows us to impute the unobserved productivity innovations for treated firms using the distribution $G_\epsilon^0(\cdot)$. In order to implement this assumption, however, we require an untreated group-time pair (g', s) such that the marginal distribution of ϵ_{is}^0 is identified, e.g., a random group of firms that is ineligible for the treatment prior to time s .

Proposition 4.3. *Under Assumption 4.2 and 4.3, the distribution of ϵ_{it}^0 , $G_\epsilon^0(\cdot)$, is identified, and the ℓ -period-ahead CATT for group g can be identified as*

$$CATT_\ell(\omega) = \mathbb{E}[\omega_{ie_i+\ell} | \omega_{ie_i-1} = \omega] - \mathbb{E}_{(G_\epsilon^0)^\ell}[\bar{h}_0^{(\ell)}(\omega_{ie_i-1}, \epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0) | \omega_{ie_i-1} = \omega],$$

where the second expectation is taken over the joint distribution of $(\epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0)$.

Proposition 4.3 motivates a simulation-based procedure to estimate ATT. This procedure includes two steps. In the first step, the researcher can identify the distribution of ϵ_{it}^0 , i.e., $G_\epsilon^0(\cdot)$, for all firms. Then for each treated firm, starting from the ex-ante productivity ω_{ie_i-1} , we can draw ϵ_{it}^0 from G_ϵ^0 , and simulate its counterfactual productivity path in ℓ periods ahead using $\bar{h}_0^{(\ell)}(\omega_{ie_i-1}, \epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0)$. The desired ATT is simply the average difference between the realized productivity and the simulated counterfactual productivity for treated firms.

5 Discussion of the Potential Productivity Process (3)

Our approach embeds production function estimation within a dynamic potential outcome framework, offering a novel perspective in the literature. Crucially, this is not simply a rephrasing of existing methods. Whereas prior studies typically rely on structural assumptions about realized productivity, we instead impose analogous assumptions on potential productivity. This shift enhances the robustness of average treatment effect estimates, particularly in the presence of model misspecification.

As a point of comparison, consider a model based on realized productivity that allows for more flexible dynamics during the transition period. One could modify the endogenous productivity process by replacing D_{it} with the treatment history (D_{it}, D_{it-1}) in $\tilde{h}$, yielding:

$$\omega_{it} = \tilde{h}(\omega_{it-1}, D_{it}, D_{it-1}) + \epsilon_{it}. \quad (14)$$

This generalized specification appears attractive, as it directly addresses the misspecification issue illustrated in the right panel of Figure 2. By incorporating treatment history, equation (14) isolates the transition period and assigns it a distinct evolution process, $\tilde{h}(\omega, D_{it} = 1, D_{it-1} = 0)$. However, as we demonstrate in this section, this approach can still overlook important unobserved heterogeneity and yield misinterpretation of treatment effects.

Unobserved Heterogeneity Productivity plays a central role in firms' treatment decisions, especially within a dynamic optimization framework. In such settings, current productivity informs firms' expectations about future revenue streams, thereby influencing their assessment of the expected net benefits from adopting a treatment.

In the potential productivity model (3), both ω_{it}^1 and ω_{it}^0 are relevant to firms' treatment decisions, as the realized productivity—and thus future revenues—depend on the treatment status. Although firms are assumed to observe both potential productivities when making adoption choices, econometricians can typically recover only ω_{it}^0 prior to treatment. Consequently, model (3) is capable of capturing unobserved heterogeneity in firms' potential productivity paths.

In contrast, under the model specified in equation (14), only the realized productivity ω_{it} enters firms' treatment decisions, as future productivity is assumed to be forecastable solely based on the observed state variables ω_{it} . Once the realized productivity process is recovered, the econometrician possesses the same information set as the firm, implying that no unobserved heterogeneity in productivity remains in this framework.

Causal Interpretation Ignoring unobserved heterogeneity can lead to misinterpreting the switching-period dynamics as treatment effects. Although the generalized productivity process in equation (14) provides a structural interpretation of productivity evolution during the transition period, it implicitly assumes that the $\tilde{h}(\omega_{it-1}, D_{it} = 1, D_{it-1} = 0)$ estimated from the subsample of treatment-switching firms represents the transition dynamics for all firms: given the estimated $\tilde{h}$, we can also counterfactually simulate the ω_{it}^1 for an untreated firm if the firm had taken treatment in period t . However, if selection into treatment is driven by unobserved differences in potential productivity, the interpretation of the estimated $\tilde{h}$ cannot be generalized to the untreated firms due to the selection bias. In contrast, our approach in equation (3) remains agnostic about the switching-period process, thereby avoiding stronger assumptions that lead to incorrect causal inferences when these assumptions are violated.

Misleading Interpretation of Treatment Effects The generalized productivity process can yield misleading estimates of the average treatment effect (ATE) by implicitly relying on a conditional unconfoundedness assumption—namely, that $\omega_{it}^1, \omega_{it}^0 \perp D_{it} \mid \omega_{it-1}$.

However, if the true productivity process follows equation (3), this unconfoundedness assumption fails at the treatment switching period. Specifically, although $\omega_{it-1} = \omega_{it-1}^0$ is observed prior to treatment, we cannot control for the unobserved heterogeneity embedded in ω_{it}^1 , nor its potential correlation with D_{it} .¹¹

In contrast, the productivity process defined by equation (14) assumes away this source of heterogeneity by treating ω_{it}^1 and ω_{it}^0 as conditionally independent of D_{it} given ω_{it-1} . Under this specification, all variation in ω_{it} arises from the innovation term ϵ_{it} , which is assumed to be mean-independent of D_{it} by Assumption 2.3. This assumption effectively rules out any selection on ω_{it-1}^1 , leading to a falsely identified ATE.

The key source of bias comes from the wrong interpretation of $\tilde{h}(\omega_{it-1}, D_{it} = 1, D_{it-1} = 0)$, which is used to generate counterfactual treated outcomes for untreated firms. If this function is biased due to selection on unobservables such as ω_{it-1}^1 , then the imputed values of ω_{it}^1 for untreated firms will also be biased, resulting in a misleading estimate of the ATE.

6 Empirical Study

6.1 Background

The rise of technologies such as artificial intelligence, robotics, cloud computing and big data analytics, has ushered in a new era of digitalization in firms' production activities. This transformation has sparked significant interest among researchers and policymakers due to its potential for productivity growth. In this section, we apply our proposed method using a firm-level dataset from China to investigate the impact of manufacturing firms' production digitalization on productivity growth.

Production digitalization refers to the integration of advanced digital technologies and tools throughout the entire production process. For a long time, China's manufacturing has concentrated on making low-end, labor-intensive goods. Against this backdrop, the Chinese government has been intensively investing in infrastructure in information and communication technologies. In 2015, China issued the *Made in China 2025* as a national development plan and a comprehensive set of industrial policies to further de-

¹¹The treatment decision D_{it} may be correlated with prior productivity ω_{it-1} , which in turn is dynamically related to ω_{it} via the potential productivity process.

velop China’s manufacturing sector. In response to these efforts, Chinese manufacturing firms have been actively investing in digital technologies to upgrade their production processes. We believe this application exemplifies the added generality of our methodology because productivity after digitalization likely depends on intangible knowledge and ability that is not well proxied by total factor productivity prior to digitalization.

6.2 Data

The empirical study combines two datasets. The first dataset covers publicly traded manufacturing firms in the Chinese stock market between 2005 and 2019. This dataset is collected by CSMAR (similar to Compustat in the US) and contains rich information on firms’ production activities. The second dataset is the annual reports for China’s A-shares manufacturing firms downloaded from the websites of the Shanghai Stock Exchange, Shenzhen Stock Exchange, and CNINF¹² between 2005 and 2019. We use the texts in the annual reports to construct the variable of production digitalization. Our data covers the period in which many Chinese firms started to adopt production digitalization as an important development strategy. Due to the popularity of digitalization among stakeholders and policymakers, we believe that firms have adequate incentive to record any digitalization of their production process in their annual reports to shareholders.

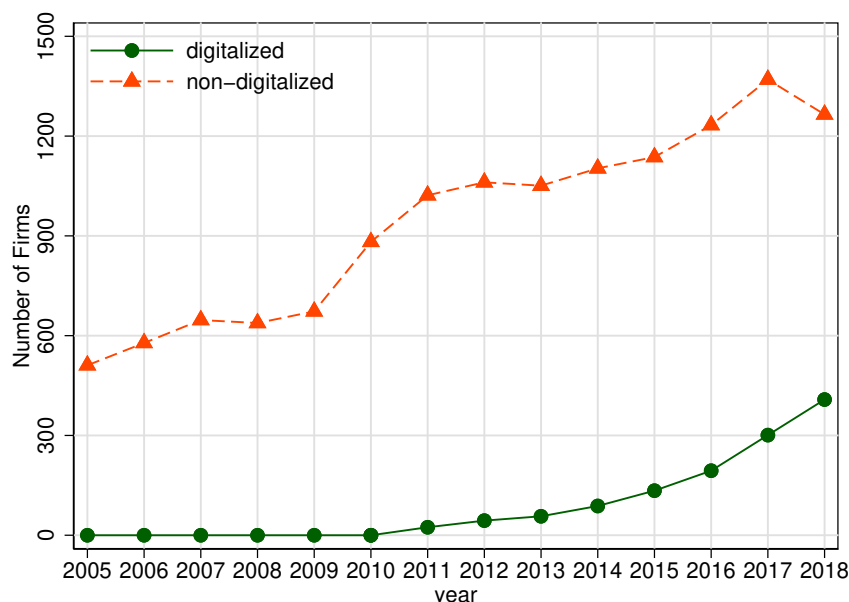
We construct the measure of production digitalization by combining text analysis tools with manual reading of the annual reports of listed manufacturing firms. Based on a set of digitalization-related technologies (e.g., big data analytics, artificial intelligence, internet of things (IoT), cloud computing, and robotics), we first extract the digitalization-related keywords and manually read the texts around the keywords in each annual report. Following Zhai et al. (2022), we hired two research assistants independently to manually read the extracted texts to determine whether the firm undertakes production digitalization each year. In particular, as an improvement of the existing method, we have excluded scenarios in which the firm only describes the development of digitalization as a trend in its own industry or as an introduction of the national development strategy. The detailed procedures and several concrete examples of texts on the identified production digitalization are presented in Appendix E. For empirical purposes, we define a dummy variable Digit_{it} to capture the status of production digitalization of firm i in year t . The variable $\text{Digit}_{it} = 1$ for years after year t if firm i ’s initial year of production digitalization is identified as year t . Otherwise, $\text{Digit}_{it} = 0$. Note that by the construction of Digit_{it} , the

¹²CNINF (<http://www.cninfo.com.cn/new/index>) is a large-scale professional website that discloses the announcement information and market data of more than 2500 listed companies in Shenzhen and Shanghai.

treatment is absorbing, fitting the context of our econometric framework.¹³

Figure 3 shows the strong growth of production digitalization within the sample period in the raw data. The number of firms that have adopted production digitalization was zero before 2011, but increased rapidly to 408 in 2018. This is consistent with the rapid development of the digital economy and the building of infrastructure for information technologies in China during this period. Note that there has been an increase in the number of non-digitalized firms. This is because more manufacturing firms were listed during the sample period, and firm exit is relatively rare. The growing trend of production digitalization in our sample provides a suitable empirical setting to employ our proposed method to investigate the productivity effects of production digitalization.

Figure 3: Trend of Production Digitalization



Note: Based on the authors' calculation using the raw sample.

To account for industrial heterogeneity and keep a sufficient number of observations in each industry, we classify firms according to the first digit of the industry code. We require that the number of treated observations is no less than 50 for each industry. Our final estimation sample contains six main industries with 9717 observations (8916 untreated and 801 treated). We provide details on the variable definitions, data cleaning procedures and the summary statistics in Appendix E.

¹³One may argue that the level of digitalization may change over time. We acknowledge this limitation and leave the investigation of the dynamic treatment effect for continuous treatment variables as an important future research direction.

6.3 Identification Assumptions and Estimation Procedures

6.3.1 Identification Assumptions

Our identification strategy relies on the conditional parallel trends (Assumption 4.1), which asserts that, if the firm had not started production digitalization, its counterfactual productivity would have continued to evolve according to $\bar{h}_0$. However, we emphasize that, because we assume the adoption of digital technologies is an absorbing treatment state, the conditional parallel trend assumption is untestable. We cannot observe the evolution of counterfactual productivity once a firm starts production digitalization, and the counterfactual productivity is not relevant to any future actions or outcomes of the firm.

In the present application, we also invoke Assumption 4.3 to identify the treatment effect of production digitalization on productivity. This assumption requires that, for each treated firm, there exists a comparison group that shares the same distribution of untreated productivity innovations ϵ_{it}^0 , conditional on the firm's pre-adoption productivity level. We use observations from the three years immediately preceding each industry's initial year of digitalization adoption as the comparison group, which can mitigate the potential bias from long-term structural changes and account for industry heterogeneity in initial adoption time.

To provide evidence of the validity of Assumption 4.3, we compare the distribution of productivity innovations ϵ_{it}^0 for treated and control groups during three years prior to treatment adoption. Appendix E.3 presents the supporting evidence: In Figure E.1, we plot the cumulative distribution functions (CDFs) of ϵ_{it}^0 for both groups across industries, and there is no visually distinguishable difference between the CDFs for the two groups. We conduct a Kolmogorov-Smirnov test of the distributional differences in Table E.3, and we fail to reject the null hypothesis that the two CDFs of ϵ_{it}^0 for the two groups are the same at the 5% significance level for all industries.

6.3.2 Production Function Estimation

We employ the Akerberg et al. (2015) method to estimate a value-added production function, with the extension of the potential productivity process (3). We use the translog specification as the benchmark model:

$$y_{it} = \beta_t t + \beta_l l_{it} + \beta_k k_{it} + \beta_{ll} l_{it}^2 + \beta_{kk} k_{it}^2 + \beta_{lk} k_{it} l_{it} + \omega_{it} + \eta_{it}, \quad (15)$$

where y_{it} , l_{it} , k_{it} are the logged value-added, logged number of employees, and logged capital, respectively. Any exogenous trends in the productivity are captured by $\beta_t t$, and

η_{it} is an exogenous idiosyncratic ex-post output shock. As we have mentioned, China has been actively investing in infrastructure of information and communication technologies with the goal of upgrading the manufacturing sector. The exogenous time trend in the production function may capture the common trend in productivity growth. In light of our econometric framework, the realized productivity ω_{it} can be expressed as $\omega_{it} = \text{Digit}_{it} \times \omega_{it}^1 + (1 - \text{Digit}_{it})\omega_{it}^0$, where $\text{Digit}_{it} \in \{0, 1\}$ is the defined indicator for production digitalization. We assume $\bar{h}_0$ and $\bar{h}_1$ are cubic polynomials of the potential productivity with and without digitalization:

$$\omega_{it}^d = \rho_0^d + \rho_1^d \omega_{it-1}^d + \rho_2^d (\omega_{it-1}^d)^2 + \rho_3^d (\omega_{it-1}^d)^3 + \epsilon_{it}^d, \quad d \in \{0, 1\}, \quad (16)$$

where $d = 1$ indicates the treated firms that have adopted production digitalization in the sample period, and $d = 0$ represents the control firms that have never started production digitalization in the sample.

Because the productivity process in the period of adoption cannot be modeled by (16), we drop this period when estimating the production function.¹⁴ Guided by Theorem 3.1, we construct moment conditions using the instruments of Akerberg et al. (2015). To account for the industrial heterogeneity in production technologies, we estimate the production functions and productivity evolution processes separately for each industry. The estimation results for the production functions are presented in Table E.4. Notably, the estimation results show a significant positive time trend in the production function, indicating rapid technological progress in the Chinese manufacturing sector. This is also consistent with China's intensive investment in infrastructure and information and communication technologies.

6.3.3 Estimation of the Effects of Production Digitalization on Productivity

After the estimation of the production function, we compute the productivity and recover the productivity evolution process. Based on the productivity estimates and the recovered productivity process, we use the proposed simulation-based approach to estimate the firm-specific treatment effects by constructing multiple counterfactual productivity paths for each firm. To simulate counterfactual productivity paths for treated units, we draw productivity innovations ϵ_{it}^0 from the observations within three years before the initial treatment adoption time in each industry.

Following the identification argument for the CATT and the simulation-based method

¹⁴An alternative way is to add a dummy variable indicating the transition period. Our results are robust to this alternative specification.

in Proposition 4.3, we propose to study the following firm-specific ℓ -period treatment effect $TT_{i\ell} \equiv \omega_{ie_i+\ell} - E_{G_\epsilon^0}[\bar{h}_0^{(\ell)}(\omega_{ie_i-1}, \epsilon_{e_i}^0, \dots, \epsilon_{e_i+\ell}^0)]$. The $TT_{i\ell}$ is a firm-specific treatment effect averaged over all counterfactual paths of $\omega_{ie_i+\ell}^0$. If we average the $TT_{i\ell}$ for all firms such that $\omega_{ie_i-1} = \omega$, then such an average, by Proposition 4.3, will be the $CATT_\ell(\omega)$. This firm-specific object allows us to study the heterogeneity of treatment effect across firms that would be missed in the aggregated ATT estimates. Moreover, the $TT_{i\ell}$ is easier to calculate than the CATT, which requires taking averages of $TT_{i\ell}$ across firms that have the same lagged productivity, and the mean of $TT_{i\ell}$ is the ATT_ℓ . Specifically, for firm i that started production digitalization in year e_i , we estimate the firm-specific ℓ -period treatment effects of production digitalization as

$$\widehat{TT}_{i\ell} = \hat{\omega}_{ie_i+\ell} - \frac{1}{M} \sum_{m=1}^M \hat{\omega}_{ie_i+\ell}^0(m), \quad (17)$$

where $\hat{\omega}_{ie_i+\ell}^0(m)$ is the unrealized potential productivity obtained through the simulated productivity path m , and M is the total number of counterfactual productivity paths. In our estimation, we set M to be 100. After experimentation, we noticed that the TT estimate is sensitive to the outliers in the distribution of potential productivity innovations ϵ_{it}^0 . To deal with this problem, instead of drawing from the nonparametric distribution of productivity innovations, we exclude the outliers of productivity innovations by discarding values smaller than the 1st percentile or greater than the 99th percentile and assume a Gaussian distribution for the productivity innovations: $\epsilon_{it}^0 \sim \mathcal{N}(0, \sigma_\epsilon^2)$. The standard deviation σ_ϵ is estimated as the sample analog.

Based on the estimated firm-specific treatment effects, we then compute group-specific treatment effects. We consider two types of group-specific treatment effects: the first is the dynamic treatment effects, which are obtained by averaging $\widehat{TT}_{i\ell}$ by period ℓ ; the second is the industrial treatment effects, which are computed by averaging $\widehat{TT}_{i\ell}$ by industries.

Due to the concerns about an overly small sample size of the treated firms, we set ℓ to be 0 to 3. We use the block-bootstrap to construct the confidence interval for the group-level ATT estimates. In particular, we resample observations for each industry by firm-level clusters and repeat the two-step estimation procedure. Considering that different industries have distinct levels of production digitalization, we also bootstrap the sample by industry level.¹⁵

¹⁵The estimation is implemented in Stata, and the code is extended from the `prodest` package developed by Rovigatti and Mollisi (2018). We thank Gabriele Rovigatti for sharing the code.

6.4 Empirical Results

6.4.1 Group-specific Average Treatment Effects

Dynamic Average Treatment Effects We first present the estimation results for the dynamic treatment effects in Table 1.¹⁶ We find that production digitalization is associated with positive productivity effects in the first two periods (periods 0 and 1), followed by slightly negative effects in periods 2 and 3. On average, the estimated effect of production digitalization on productivity is approximately -0.001. Importantly, none of these estimates are statistically significant at the 10% level, indicating that, on average, production digitalization has not led to significant productivity gains among Chinese manufacturing firms. In Section 6.4.2, we further investigate the heterogeneity in firm-level treatment effects.

Table 1: Treatment Effects on Productivity

Periods	ATT	SE	Treated Obs.
0	0.012	0.011	271
1	0.006	0.017	187
2	-0.015	0.024	124
3	-0.038	0.033	85
Total	-0.001	0.015	667

Note: The production function is specified as the translog production function. For each firm, 100 counterfactual productivity paths are simulated. Standard errors are obtained by bootstrapping 500 times.

Industrial Average Treatment Effects Table 2 reports the industry-level treatment effects and their contributions to the overall treatment effect in the sample. The overall causal effect of production digitalization is obtained by averaging across all observations. Due to limited sample sizes within each industry, we do not report dynamic treatment effects at the industry level. Electronics manufacturing ($\widehat{ATT} = 0.085$), healthcare manufacturing ($\widehat{ATT} = 0.026$), and equipment manufacturing ($\widehat{ATT} = 0.002$) exhibit positive treatment effects on productivity, contributing approximately 595.0%, 151.3%, and 134.3% to the sample's overall ATT, respectively. Among these, the effect for electronics manufacturing is statistically significant at the 10% level. The remaining three industries show negative treatment effects on productivity. Overall, the average treatment effect is slightly negative and close to zero. This industry-level heterogeneity suggests that firms

¹⁶The estimated parameters for the translog production function are reported in Appendix E.4.

experience differential productivity gains from adopting production digitalization. The relatively larger positive effects observed in industries such as electronics, healthcare, and equipment manufacturing may reflect their more complex production processes and higher technological intensity, where the adoption of digital technologies can generate substantial efficiency improvements, enhance precision, and enable greater customization. In contrast, industries with simpler production processes may derive more limited productivity benefits from digitalization.

Table 2: Industry-level Treatment Effects on Productivity

Industries	Mean	SE	Contribution (%)	Treated Firms
Electronics Manufacturing	0.085*	0.044	595.0	48
Healthcare Manufacturing	0.026	0.047	151.3	40
Equipment Manufacturing	0.002	0.019	134.3	402
Chemical Synthesis	-0.033	0.049	-321.0	66
Metal Processing	-0.041	0.032	-340.2	57
Food & Beverage	-0.041	0.074	-319.3	54
Total	-0.001	0.015	-100.0	667

Note: The contribution of each industry is calculated as the ratio of sample-share-weighted treatment effects to the average treatment effects in the whole sample. * $p < 0.10$

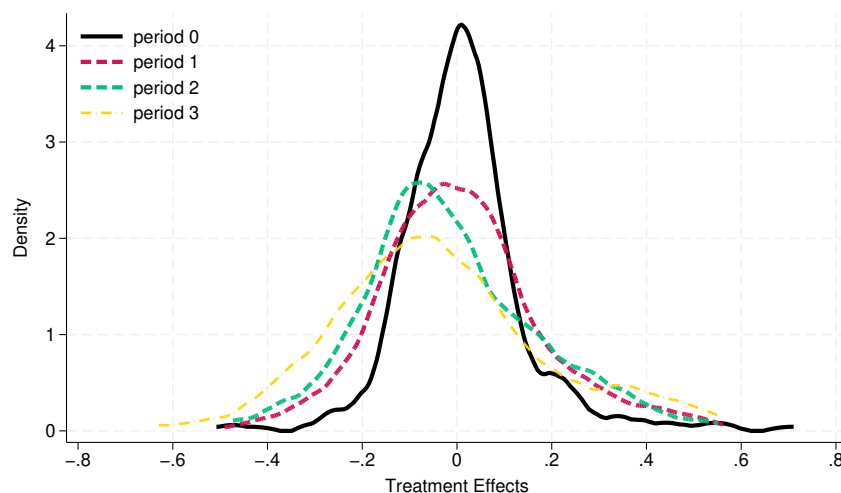
6.4.2 Firm-specific Treatment Effects

In Figure 4, we display the kernel density of the firm-specific treatment effects in different periods after production digitalization. The large variation in productivity gains may reflect the differences in firms' organizational efficiency in building the new digital production technology and the learning ability to harness the new digital technology in the production process.

The density of firm-specific treatment effects ($\widehat{\text{TT}}_{it}$) is more dispersed and appears to shift to the left from period zero ($\ell = 0$) to three periods after digitalization. This indicates that production digitalization tends to have smaller or more negative productivity effects over time for many firms. However, the increased dispersion indicates that firms' experiences with production digitalization also become more strongly differentiated. This is consistent with the observation that the success rate of digital transformation is low (Bughin et al., 2019), and also largely supports the theory that firms may encounter organizational or technological barriers in the process of upgrading their business practices and the skills of the workforce in order to fully harness the new technology (Taylor and Helfat, 2009; Feigenbaum and Gross, 2021). However, as production digitalization has large negative productivity effects for a non-negligible portion of firms, the mean of digi-

talization on productivity remains negative in later periods.

Figure 4: Firm-specific Treatment Effects of Production Digitalization on Productivity



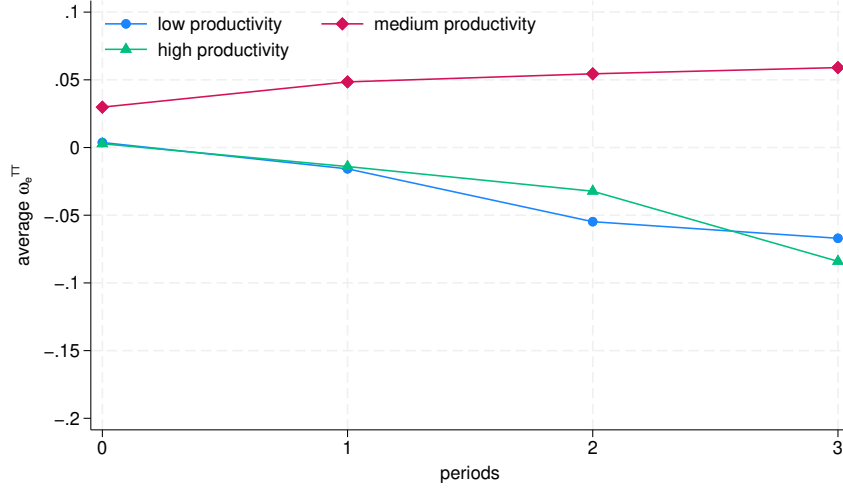
Note: This figure shows the probability density of firm-specific treatment effects of digitalization on productivity. Firm-specific treatment effects on productivity are obtained by simulating 100 counterfactual productivity paths for each treated observation.

Next, in Figure 5, we investigate the CATT for different levels of productivity prior to production digitalization. To ensure that we have enough sample size, for each industry, we construct three productivity bins by splitting the productivity evenly into three groups based on the percentiles of the productivity. As time evolves, the productivity growth caused by production digitalization of the low- and high-productivity firms changes from positive to negative. In contrast, the productivity gains for medium-productivity firms tend to be larger in all periods than low-productivity firms.

These heterogeneous patterns are broadly consistent with prior literature emphasizing the challenges firms face in fully capturing the benefits of digital technologies. Several studies have documented that adopting advanced digital tools often requires substantial complementary investments in organizational capabilities, human capital, and process re-engineering (Brynjolfsson and McAfee, 2014; Bresnahan et al., 2002; Tambe et al., 2012). Low-productivity firms may lack the managerial capacity, skilled workforce, or financial resources necessary to fully exploit the potential of digitalization, leading to limited or even negative long-run productivity effects. Conversely, highly productive firms may already operate close to technological frontiers, leaving less room for additional gains and potentially exposing them to coordination or complexity costs associated with integrating new technologies (Bloom et al., 2020; Aghion et al., 2017). Medium-productivity firms

may occupy a “sweet spot,” where the adoption of digital technologies offers meaningful opportunities for efficiency improvements and remains within the firm’s absorptive and managerial capacity. Our results therefore contribute to a growing body of evidence highlighting that the productivity returns to digitalization are highly heterogeneous and contingent on firm capabilities.

Figure 5: Initial Productivity and the Dynamic Effects of Digitalization on Productivity



Note: The horizontal axis indicates the periods after production digitalization. The initial productivity is normalized by subtracting the industry-level average productivity to facilitate the cross-industry comparison.

6.4.3 Comparison with Ex-post Regressions

To emphasize the difference between our method and the existing methods, we also estimate the treatment effects on productivity using the ex-post regression method. We focus on the electronic manufacturing industry because there is a positive impact of production digitalization on productivity. We estimate the following two-way fixed effects model:

$$\hat{\omega}_{it} = \delta \text{Digit}_{it} + \rho_1 \hat{\omega}_{it-1} + \rho_2 \hat{\omega}_{it-1}^2 + \rho_3 \hat{\omega}_{it-1}^3 + \lambda_i + \lambda_t + u_{it}, \quad (18)$$

where $\hat{\omega}_{it}$ is the productivity estimate for firm i in year t , and Digit_{it} is the dummy variable indicating production digitalization. The parameters λ_i and λ_t represent the firm and year fixed effects, respectively. The error term is u_{it} . The parameter δ is usually interpreted as the treatment effect of production digitalization on productivity (e.g., [Liu and Mao \(2019\)](#)).

We follow the estimation strategy of an ex-post method to estimate the productivity and then run the regression as specified in equation (18). Moreover, we experiment with

three ways of estimating the production function and productivity. The first productivity process we specify is an exogenous productivity process without including the information on digitalization. Specifically, the productivity process is as follows:

$$\omega_{it} = \varrho_0 + \varrho_1\omega_{it-1} + \varrho_2\omega_{it-1}^2 + \varrho_3\omega_{it-1}^3 + \varepsilon_{it}. \quad (19)$$

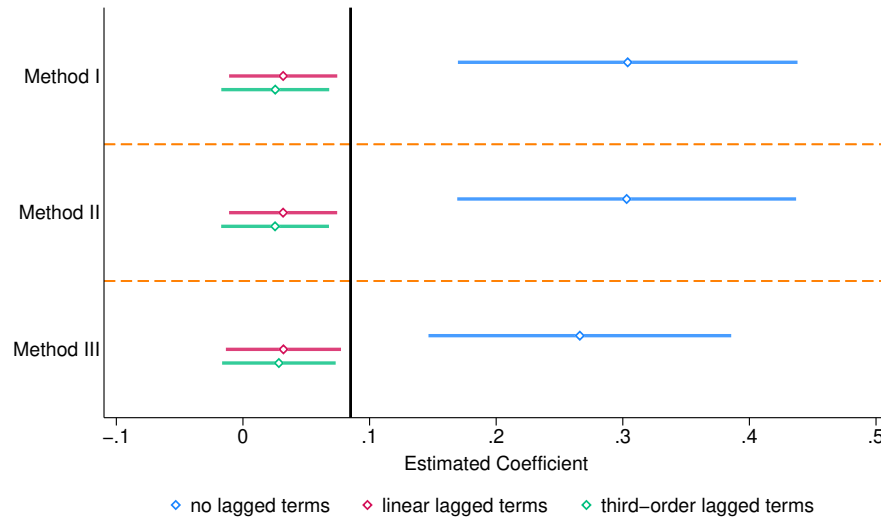
In the other two productivity processes, we specify an endogenous productivity process by including the variable of production digitalization in the productivity process:

$$\begin{aligned} \omega_{it} = & \tilde{\varrho}_0 + \tilde{\varrho}_1\omega_{it-1} + \tilde{\varrho}_2\omega_{it-1}^2 + \tilde{\varrho}_3\omega_{it-1}^3 \\ & + (\tilde{\varrho}'_1\omega_{it-1} + \tilde{\varrho}'_2\omega_{it-1}^2 + \tilde{\varrho}'_3\omega_{it-1}^3) \times \text{Digit}_{it} + \tilde{\delta}\text{Digit}_{it} + \epsilon_{it}, \end{aligned} \quad (20)$$

and estimate the production function either using the entire sample or excluding the transition period. After obtaining the productivity estimates, we then estimate equation (18), experimenting with controlling for different lagged productivity terms.

We summarize the coefficient estimates for the treatment indicator of production digitalization in Figure 6, with the full regression results reported in Table E.5. The vertical black line indicates the average treatment effect estimated by our method, which is 0.085. The estimated coefficients on Digit_{it} are positive across the various existing methods, though their magnitudes and significance levels vary depending on the specific method and model specification. Notably, the coefficient estimates obtained from simple regressions of productivity on the treatment indicator, which is represented by the diamond markers on the blue horizontal lines, tend to be larger than our estimated average treatment effect. This discrepancy likely reflects an upward bias in the simple regressions, as they do not account for the potential endogeneity arising from the selection into treatment based on unobserved potential productivity ω_{it}^1 before treatment: more productive firms may be more likely to adopt production digitalization, leading to overestimated treatment effects when this selection is not properly addressed. In addition, the coefficient estimates from models that include lagged productivity terms (represented by the red and green horizontal lines) are generally smaller than those from the simple regressions. This pattern may reflect the dynamic nature of productivity: the contemporaneous effects of production digitalization may be partially absorbed by the lagged dependent variable, as productivity gains in one period carry over into subsequent periods. As a result, these specifications may underestimate the total treatment effect if the dynamic transmission mechanism is not fully captured.

Figure 6: Comparison between Our Method and Other Methods



Note: “Method I” is the ex-post regression method with exogenous productivity process (19), “Method II” is the ex-post regression method with endogenous productivity process (20) using the full sample, and “Method III” is the ex-post regression method with endogenous productivity process (20) excluding the transition period. The solid horizontal lines represent the 95% confidence intervals, with blue for regression without lagged productivity, red for regression with lagged first-order productivity, and green for regression with lagged first-, second- and third-order productivity. The black vertical line represents the average treatment effect estimated by our method.

7 Conclusion

This article generalizes standard production function estimation methods by incorporating a binary treatment that affects the evolution of firm-level productivity and production technology. As a theoretical framework, this generalization illuminates potential issues when we identify the parameters of a model with both structural and reduced-form features. As a practical tool, our framework suggests a robust method of estimating the effect of treatment on firm-level productivity that only makes the necessary assumptions to identify the treatment effect and avoids common specification issues with regression-based approaches.

Equivalently, our framework can be viewed as a generalization of the dynamic potential outcome framework to a setting in which the outcome is not directly observed, but must be inferred through the lens of a model. In our case, the outcome of interest is the total factor productivity, which must be inferred under additional assumptions about firm behavior. From either perspective, the goal of the framework is to reconcile the sets of assumptions required to infer the productivity and identify the treatment effect on it.

The discussion in this article exclusively pertains to treatment effects on total factor productivity because this setting describes much of the recent literature, but our approach generalizes to other settings. We leave these alternatives and extensions for future work. Some natural alternatives would involve estimating total factor productivity using other means, for instance, using cost shares to estimate input elasticities of output and extracting the residual from the logged output regression as a measure of log-productivity. Compared with our baseline model, this alternative makes additional assumptions about the firm's optimizing behavior and relaxes assumptions about the evolution of productivity. It could be implemented in the same spirit as our approach by separately estimating cost shares in the treated and untreated states, although some assumptions would be required to estimate the missing counterfactual when treatment is not exogenous. In this article, the Markov assumption on the potential productivity process provides additional moments to estimate the production function as well as permits matching treated and untreated firms on the basis of their lagged productivities. Without assuming a Markov productivity process, some other assumptions would have to fulfill this second role.

Other alternatives might consider factor-augmenting productivity instead of, or in addition to, total factor productivity. Unless the alternative productivity measure is directly measured, such as value-added per hour of labor, these alternatives would most likely entail adaptations of the moment conditions used to recover the outcome. Again, if the new productivity measure is not assumed to be Markov, another assumption would have

to be invoked to estimate the missing counterfactual when treatment is endogenously assigned.

Extensions such as allowing for multiple levels of treatment could also be routinely incorporated by applying our binary treatment approach to pairwise differences in the levels of treatment. Other extensions are less obvious but no less desirable in some circumstances, such as accommodating a continuous treatment variable.

As an example of an application of our baseline methodology, we ask whether firm-level data can explain the apparent lack of a productivity revolution in aggregate productivity measures following the production digitalization. Consistent with the macro-level data, we find positive but statistically insignificant effects of production digitalization on productivity. However, our analysis uncovers substantial heterogeneity across firms and industries, as well as over time. Specifically, firms with ex-ante medium productivity tend to enjoy greater productivity gains three years after adopting new digital technologies. In sharp contrast, we find biased estimates of the effects of digitalization on productivity using regression-based methods that mimic existing approaches.

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Appendices

A Connection to the Dynamic Treatment Effect

A.1 The No-Anticipation and Sequential Randomization Condition

We now briefly connect our method to the dynamic treatment effect literature (Abbring and Heckman, 2007). There are two key conditions in the dynamic treatment effect literature: the no-anticipation condition (NA) and the sequential randomization condition (SR). Because our framework combines both the potential outcome model and the structural equation model, we can use the structural model to verify whether NA and SR conditions hold or not. Following the notation in Abbring and Heckman (2007), we let $\mathbf{D}_i^t = (D_{i1}, \dots, D_{it})$, and $\omega_i^{dt} = (\omega_{i1}^d, \dots, \omega_{it}^d)$ for $d = 0, 1$. We state the NA condition in our framework.

Assumption A.1. (NA) Let $\mathbf{D}_i^T$ and $\tilde{\mathbf{D}}_i^T$ be two treatment sequences such that $\mathbf{D}_i^t = \tilde{\mathbf{D}}_i^t$ for any $t \leq T$. The no-anticipation condition holds if the potential $(\omega_{it}^0, \omega_{it}^1)$ generated under $\mathbf{D}_i^T$ coincides with the potential $(\tilde{\omega}_{it}^0, \tilde{\omega}_{it}^1)$ generated under $\tilde{\mathbf{D}}_i^T$ for all $t \leq T$.

The no-anticipation condition says that if two sequences of treatment coincide up to time t , then the potential productivity up to time t should also coincide. No-anticipation is the crucial assumption for the analysis of dynamic treatment effects (Sun and Abraham, 2021).

Given the Markovian evolution process (3), NA Assumption A.1 holds as long as there is no anticipation in the productivity innovations: the innovation sequence $(\epsilon_{is}^0, \epsilon_{is}^1)_{s \leq t}$ under $\mathbf{D}_i^t$ coincides with the innovation sequence $(\tilde{\epsilon}_{is}^0, \tilde{\epsilon}_{is}^1)_{s \leq t}$ under $\tilde{\mathbf{D}}_i^t$. We view Assumption A.1 as a weak requirement as the innovations to the productivity process are usually assumed to be unexpected by firms in the productivity estimation literature.

Another condition is the sequential randomization condition (Robins, 1997; Gill and Robins, 2001; Lok, 2008), which says that future potential outcomes are conditionally independent of the current treatment status. Sequential randomization is crucial to the identification of average treatment effects. We state the firm's SR condition in our framework.

Assumption A.2. (SR-F) $D_{it+1} \perp (\omega_{is}^1, \omega_{is}^0)_{s \geq t} | \mathcal{I}_{it}^F$ holds for all t , where $\mathcal{I}_{it}^F = \mathcal{Z}_{it} \cup \{\omega_{is}^1, \omega_{is}^0\}_{s \leq t-1}$ is the firm's information set.

We call Assumption A.2 the sequential randomization condition for firms because we condition on the firms' information. This is slightly different from the traditional sequential randomization condition in Gill and Robins (2001), where their condition is imposed on the econometrician's information set: $\mathcal{I}_{it}^E \equiv \mathcal{Z}_{it} \cup \{\omega_{is}\}_{s \leq t-1}$.

Our structural model implies that Assumption A.2 holds. Indeed, from the firm's dynamic optimization behavior, we know D_{it+1} is a function of $\mathcal{I}_{it}^F$, denoted by $D_{it+1} = g(\mathcal{I}_{it}^F)$. Then given the information set $\mathcal{I}_{it}^F$, D_{it+1} is a degenerate variable, and thus Assumption A.2 holds. When the treatment variable is externally imposed, and the assigner randomizes the treatment up to the firm's knowledge, i.e., $D_{it+1} = \tilde{g}(\mathcal{I}_{it}^F, \eta_{it})$ for some η_{it} independent of $(\omega_{is}^1, \omega_{is}^0)_{s \geq t}$, then SR-F also holds.

Now, suppose the treatment is absorbing, and the firms can only choose the treatment status D_{ie} at a uniform time e . Under Assumption A.2, define the propensity score as $\kappa(\mathcal{I}_{ie-1}^F) \equiv E[D_{ie} | \mathcal{I}_{ie-1}^F]$. Then we can rewrite the average treatment effect as:

$$E[\omega_{ie}^1 - \omega_{ie}^0] = \mathbb{E} \left[\frac{\omega_{ie} D_{ie}}{\kappa(\mathcal{I}_{ie-1}^F)} - \frac{\omega_{ie}(1 - D_{ie})}{1 - \kappa(\mathcal{I}_{ie-1}^F)} \right]. \quad (\text{A.1})$$

In general, when the sequential randomization fails, the average treatment effect is not identified without further restrictions, see Abbring and Heckman (2007) for discussion.

A.2 Identify the Average Treatment Effect

When we write down the average treatment effect equation (A.1), we use the firms' information set $\mathcal{I}_{ie-1}^F$. Many variables in $\mathcal{I}_{ie-1}^F$, such as $(\omega_{ie-1}^1, \omega_{ie-1}^0)$, are not available to the econometrician. Instead, the econometrician has access to the information set $\mathcal{I}_{ie-1}^E$. To identify the ATE under the absorbing treatment context, we require a sequential randomization condition for the econometrician:

Assumption A.3. (SR-E) $D_{it} \perp (\omega_{is}^1, \omega_{is}^0)_{s \geq t} | \mathcal{I}_{it}^E$ holds for $t = e$.

In general, we have $\mathcal{I}_{ie-1}^F \setminus \mathcal{I}_{ie-1}^E = \{(\omega_{is}^1)_{s \leq e-1}\}$. If D_{ie} is dependent of ω_{ie-1}^1 , then Assumption A.3 fails because ω_{ie-1}^1 is dependent of ω_{ie}^1 . However, there are special cases where we can still use the econometrician's information set to identify the ATE.

Proposition A.1. Suppose the potential productivity process satisfies Example 2, i.e., $\omega_{is}^1 = \omega_{is}^0$ for $s \leq e - 1$. Then, we can identify the ℓ -period-ahead average treatment effect as

$$E[\omega_{ie+\ell}^1 - \omega_{ie+\ell}^0] = \mathbb{E} \left[\frac{\omega_{ie+\ell} D_{ie}}{\kappa(\mathcal{I}_{ie-1}^E)} - \frac{\omega_{ie+\ell}(1 - D_{ie})}{1 - \kappa(\mathcal{I}_{ie-1}^E)} \right]. \quad (\text{A.2})$$

Proof. If $\omega_{is}^1 = \omega_{is}^0$ for $s \leq e - 1$, then firms' treatment decisions satisfy $D_{ie} = g(\mathcal{I}_{ie-1}^E)$. By the potential productivity process (3), $\omega_{ie+\ell}^1 = \bar{h}_1^{(\ell)}(h_1^+(\omega_{ie-1}) + \epsilon_{ie}^0, \epsilon_{ie+1}^1, \dots, \epsilon_{ie+\ell}^1)$, and $\omega_{ie+\ell}^0 = \bar{h}_0^{(\ell+1)}(\omega_{ie-1}, \epsilon_{ie}^0, \epsilon_{ie+1}^0, \dots, \epsilon_{ie+\ell}^0)$, where the $\bar{h}_d^{(\ell)}$ is the ℓ -times composed evolution process of $\bar{h}_d$. Then conditional on the econometrician's information set $\mathcal{I}_{ie-1}^E$, the variation of $(\omega_{ie+\ell}^1, \omega_{ie+\ell}^0)$ is caused by $(\epsilon_{is}^1, \epsilon_{is}^0)_{s \geq e}$. By assumption, the innovation terms are mean independent of the D_{ie} . Therefore, Assumption A.3 is satisfied, and equation (A.2) follows by the propensity score matching method. $\square$

Even if the productivity process in Example 2 is the same as that in Doraszelski and Jaumandreu (2013), we note that the identified average treatment effect (A.2) is neither $E[h_1^+(\omega_{ie-1}) - \bar{h}_0(\omega_{ie-1})]$ nor $E[\bar{h}_1(\omega_{ie-1}) - \bar{h}_0(\omega_{ie-1})]$, which are usually interpreted as treatment-related effects in fully structural models. As we note, the quantity $h_1^+(\cdot) - \bar{h}_0(\cdot)$ reflects the trend difference, but it fails to account for the selection into treatment bias when treatment D_{it} is not exogenous.

B Proofs

B.1 Proofs in Section 3

B.1.1 Proof of Lemma 3.1

Proof. The proof of statement (i) is given in GNR. We use the techniques in GNR to prove statement (ii). Let $\omega_{it-1}(\beta) \equiv \Phi_{it-1}(k_{it-1}, l_{it-1}, m_{it-1}) - f_0(k_{it-1}, l_{it-1}; \beta)$. We first note that $\mathbb{E}[q_{it} | \{k_{it}, l_{it}, k_{it-1}, l_{it-1}, m_{it-1}\}] = f_0(k_{it}, l_{it}; \beta) - h(\omega_{it-1}(\beta))$. Then we have:

$$\begin{aligned} \frac{\partial \mathbb{E}[q_{it} | \{k_{it}, l_{it}, k_{it-1}, l_{it-1}, m_{it-1}\}]}{\partial k_{it}} &= \frac{\partial f(k_{it}, l_{it})}{\partial k_{it}}, \\ \frac{\partial \mathbb{E}[q_{it} | \{k_{it}, l_{it}, k_{it-1}, l_{it-1}, m_{it-1}\}]}{\partial l_{it}} &= \frac{\partial f(k_{it}, l_{it})}{\partial l_{it}}. \end{aligned}$$

Therefore, f is identified up to an additive constant by the existence of the solution to partial differential equations. $\square$

B.1.2 Proof of Theorem 3.1

Proof. We first look at equation (8), and the proof of expression (9) follows similarly. We can write

$$\begin{aligned} & \mathbb{E}[\omega_{it}(\beta) - \bar{h}_0(\omega_{it-1}(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] \\ &= \mathbb{E}[\omega_{it}^0(\beta) - \bar{h}_0(\omega_{it-1}^0(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] \\ &= \mathbb{E}[\epsilon_{it}^0 | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] = 0 \end{aligned} \tag{B.1}$$

where $\omega_{it}^0(\beta)$ denotes the potential productivity without treatment, recovered under parameter value β and $D_{it} = 0$. The first equality of (B.1) holds by the potential outcome equation and the last equality holds by Assumption 3.2. The moment condition (8) is well defined by Assumption 3.3. By Lemma 3.1, the result follows. $\square$

B.2 Proofs in Section 4

B.2.1 Proof of Corollary 4.1

Proof. Recall that from Proposition 3.1, β and the evolution process $\bar{h}_d$ are identified. Following the first step nonparametric regression method in Olley and Pakes (1996); Levinsohn and Petrin (2003); Akerberg et al. (2015), we first nonparametrically regress q_{it} on k_{it}, l_{it}, m_{it} :

$$q_{it} = \Phi(k_{it}, l_{it}, m_{it}) + \eta_{it},$$

and recover $\omega_{it} = \hat{\Phi}(k_{it}, l_{it}, m_{it}) - f(k_{it}, l_{it}, s_{it}; \hat{\beta})$. $\square$

B.2.2 Proof of Proposition 4.1

Proof. Note that by further conditional on the group $e_i = t$,

$$\begin{aligned} (\text{CATT}_0(\omega) | e_i = t) &=_{(1)} \mathbb{E}[\omega_{it} - \bar{h}_0(\omega_{it-1}^0) | e_i = t, i \in g, \omega_{ie_i-1} = \omega] \\ &=_{(2)} \mathbb{E}[\omega_{it} - \bar{h}_0(\omega_{it-1}) | e_i = t, i \in g, \omega_{ie_i-1} = \omega], \end{aligned}$$

where (1) follows by replacing ω_{it}^0 with the evolution process and using Assumptions 3.2 and 4.1, (2) follows by the potential outcome (2) and $\omega_{ie_i-1} = \omega_{ie_i-1}^0$. Further, take the expectation with respect to the treatment time e_i to get the result. $\square$

B.2.3 Proof of Proposition 4.2

Proof. By the definition of CATT:

$$\begin{aligned} \text{CATT}_{g,\ell}(\omega) &=_{(a)} \mathbb{E}[\omega_{ie_i+\ell} | \omega_{ie_i-1} = \omega] - \mathbb{E}[\bar{h}_0^{(\ell)}(\omega_{ie_i-1}, \epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0) | \omega_{ie_i-1} = \omega] \\ &=_{(b)} \mathbb{E}[\omega_{ie_i+\ell} | \omega_{ie_i-1} = \omega] - \mathbb{E}[\bar{h}_0^{(\ell)}(\omega_{is-1}, \epsilon_{is}^0, \dots, \epsilon_{is+\ell}^0) | i \in g, \omega_{is-1} = \omega] \quad (\text{B.2}) \\ &=_{(c)} \mathbb{E}[\omega_{ie_i+\ell} | \omega_{ie_i-1} = \omega] - \mathbb{E}[\omega_{is+\ell} | i \in g, \omega_{is-1} = \omega], \end{aligned}$$

where (a) follows by the potential productivity evolution process and the potential outcome equation, (b) follows by Assumptions 4.2, and (c) follows by the potential productivity evolution procedure for untreated firms, and $\omega_{is+\ell}^0 = \omega_{is+\ell}$ for untreated firms. $\square$

B.2.4 Proof of Proposition 4.3

Proof. With the identified $\bar{h}_0$ from Proposition 3.1, for any group- g' firm i at time s , we can recover its $\epsilon_{is}^0 \equiv \omega_{is} - \bar{h}_0(\omega_{is-1})$, so the distribution G_ϵ^0 is identified. By condition (ii) in Assumption 4.3, the joint distribution of $(\epsilon_{ie_i-1}, \dots, \epsilon_{ie_i+\ell})$ is identified as the product distribution $(G_\epsilon^0)^\ell$. The identification result follows by the evolution process (B.1). $\square$

B.2.5 Proof of Proposition C.3

Proof. We prove the result for the positive switching effect $ATT_{g,0}^+$, and the negative switching ATT follows similarly. Note that for regime change indicator $G_{ig} = 1$,

$$\begin{aligned} ATT_{g,0}^+ &=_{(a)} \mathbb{E}[\omega_{ig}^1 - \omega_{ig}^0 | D_{ig-1} = 0, D_{ig} = 1] \\ &=_{(b)} \mathbb{E}[\omega_{ig}^1 | D_{ig-1} = 0, D_{ig} = 1] - \mathbb{E}[\bar{h}_0(\omega_{ig-1}^0) | D_{ig-1} = 0, D_{ig} = 1] \\ &=_{(c)} \mathbb{E}[\omega_{ig} | D_{ig-1} = 0, D_{ig} = 1] - \mathbb{E}[\bar{h}_0(\omega_{ig-1}) | D_{ig-1} = 0, D_{ig} = 1], \end{aligned}$$

where (a) follows by definition, (b) follows by Assumptions 3.2 and 4.1, (c) follows by the potential outcome equation (2). $\square$

C Extensions

C.1 Treatment Dependent Production Functions

Treatments can also influence the production function. This is especially relevant when the treatment is to take a new technology that can change the relative efficiency of differ-

ent inputs. In this section, we extend the model to incorporate possible changes in the production function and discuss how to define the treatment effects on productivity.

Consider the treatment-dependent production technology

$$Q_{it} = e^{\omega_{it} + \eta_{it}} F(K_{it}, L_{it}, M_{it}, D_{it}; \beta).$$

We maintain all assumptions in Section 2-4. There are two differences compared to the main text: the moment conditions and the recovered productivity for ATT calculation. For the moment conditions in Theorem 3.1, we replace them with the following two-moment conditions:

$$\mathbb{E}[\omega_{it}(\beta, 0) - \bar{h}_0(\omega_{it-1}(\beta, 0)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] = 0, \quad (\text{C.1})$$

$$\mathbb{E}[\omega_{it}(\beta, 1) - \bar{h}_1(\omega_{it-1}(\beta, 1)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 1] = 0, \quad (\text{C.2})$$

where $\omega_{it}(\beta, d) = q_{it} - f(k_{it}, l_{it}, m_{it}, D_{it} = d; \beta)$.

Similar to Theorem 3.1, the production function $F(\cdot, D_{it} = 1; \beta)$ and $F(\cdot, D_{it} = 0; \beta)$ are identified up to a difference in the scale: If $(F(\cdot, 0; \beta), \bar{h}_0)$ and $(F(\cdot, 1; \beta), \bar{h}_1)$ satisfy the moment conditions (C.1)-(C.2), then $(e^{c_0} F(\cdot, 0; \beta), \tilde{h}_0)$ and $(e^{c_1} F(\cdot, 1; \beta), \tilde{h}_1)$ also satisfy the moment conditions (C.1)-(C.2) for $\tilde{h}_d(\omega) = \bar{h}_d(\omega - c_d)$. As a result, we cannot identify the scale of the productivity before and after the treatment. As a result, we must normalize the productivity before and after the treatment to interpret the treatment effect on productivity. We discuss two potential approaches to benchmark the productivity before and after the treatment.

Indexing number approach Following the indexing number method in Caves et al. (1982) and Brandt et al. (2012), for the post-treatment period, we define the “treated-productivity-with-fixed-technology” as:

$$\tilde{\omega}_{it}(\beta) = q_{it} - f(k_{it}, l_{it}, m_{it}, D_{it} = 0; \beta) \quad \text{for} \quad D_{it} = 1.$$

The idea is to find the hypothetical productivity $\tilde{\omega}_{it}$ that will deliver the same output under the untreated production function.¹⁷ Note that when the production function is not changing with the treatment, $\tilde{\omega}_{it}$ is the same as the $\omega_{it}(\beta)$. Estimation of the ATT can be then carried out by using $\tilde{\omega}_{it}(\beta)$.

Benchmark using the industry average inputs We suggest choosing a benchmark vector of inputs (K_0, L_0, M_0) , e.g. the mean input vector in an industry, and relating the scale

¹⁷The untreated production function becomes the benchmark for computing productivity.

of the production using $F(K_0, L_0, M_0, 0; \beta) = F(K_0, L_0, M_0, 1; \beta)$. In other words, we require that, before and after the treatment, an industry average firm will produce the same outputs given the same productivities.¹⁸ Although the choice of benchmark does affect the relative scale of the production technologies in the treated and untreated states, and therefore influences the estimate of the contemporaneous treatment effect, the econometrician can draw policy-relevant conclusions by choosing (K_0, L_0, M_0) to be representative of the industry or a group of firms within the industry. Under this normalization, we would conclude the intervention has no contemporaneous effect if it has no expected contemporaneous effect on the idiosyncratic component of a representative firm using the benchmark input vector. If a policymaker then asks whether this group of firms would instantly become more productive if every firm in the group were exogenously forced into the treated state, the answer would be no, on average.

C.2 Cohort Treatment Effect

In some empirical applications, we may want to distinguish firms that receive treatment at different times, as the treatment effect can differ between an earlier player and a late-comer. This is called the cohort treatment effect.

Suppose we want to study the treatment effect for firms who start treatment in year g , i.e., $e_i = g$. We slightly abuse the notation and use g to denote the cohort of the treatment, $g = \{i : e_i = g\}$. The ℓ -period ahead ATT and CATT for cohort g are defined as:

$$\begin{aligned} \text{CATT}_{g,\ell}(\omega) &\equiv E[\omega_{ie_i+\ell}^1 - \omega_{ie_i+\ell}^0 | e_i = g, \omega_{ie_i-1} = \omega], \\ \text{ATT}_{g,\ell} &\equiv E[\omega_{ie_i+\ell}^1 - \omega_{ie_i+\ell}^0 | e_i = g]. \end{aligned}$$

Once we identify $\text{CATT}_{g,\ell}(\omega)$, we can identify $\text{ATT}_{g,\ell} = E[\text{CATT}_{g,\ell}(\omega)]$. We replace Proposition 4.2 by

Proposition C.1. *Under Assumption 4.1, the contemporaneous CATT is identified as $\text{CATT}_{g,0}(\omega) = \mathbb{E}[\omega_{ie_i} - \bar{h}_0(\omega_{ie_i-1}) | i \in g, \omega_{ie_i-1} = \omega]$. Consequently, the 0-period-ahead ATT is identified as $\text{ATT}_{g,0} = \mathbb{E}[\omega_{ie_i} - \bar{h}_0(\omega_{ie_i-1}) | i \in g]$.*

To identify the ℓ -period ahead treatment effect, we replace Assumption 4.2 with the following assumption

Assumption C.1. *There is a group-time pair (g', s) such that (i) for all firms $i' \in g'$ that are untreated by ℓ -periods since time s , i.e., $e_{i'} > s + \ell$; and (ii) the conditional distribution of*

¹⁸This approach chooses the same benchmark for both pre- and post-treatment production functions.

$(\epsilon_{ie_i}^0, \dots, \epsilon_{ie_i+\ell}^0) | (i \in g, \omega_{ie_i-1}^0)$ is the same as the conditional distribution of $(\epsilon_{i's}^0, \dots, \epsilon_{i's+\ell}^0) | (i' \in g', \omega_{is-1}^0)$.

The group-time pair (g', s) serves as the match for the cohort- g group. The cohort dynamic treatment effect can be identified as:

Proposition C.2. *Suppose Assumption 4.1 and C.1 hold. Then the ℓ -period-ahead CATT is identified as $CATT_{g,\ell}(\omega) = \mathbb{E}[\omega_{ie_i+\ell} | i \in g, \omega_{ie_i-1} = \omega] - \mathbb{E}[\omega_{is+\ell} | i \in g', \omega_{is-1} = \omega]$. The corresponding ATT is identified as $ATT_{g,\ell} = \mathbb{E}[CATT_{g,\ell}(\omega_{ie_i-1}) | i \in g]$, where the expectation is taken over the conditional distribution of ω_{ie_i-1} given $i \in g$.*

C.3 ATT for Non-absorbing Treatment

In some scenarios, the treatment is non-absorbing by nature. In reality, firms participate in importing, exporting, or R&D activities occasionally.¹⁹ We now discuss the identification of the effects of non-absorbing treatment. As treatment can be volatile, the individual treatment effect can be influenced by a sequence of past treatment status²⁰.

Let G_{it} denote the switching period indicator such that $G_{it} = 1$ if $D_{it} = D_{it-1}$, $G_{it} = 1$ if $D_{it} > D_{it-1}$, and $G_{it} = -1$ if $D_{it} < D_{it-1}$. We expand (3) to accommodate the switch from treated to untreated case:

$$\begin{aligned} \omega_{it}^1 &= \mathbb{1}(G_{it} = 0)\bar{h}_1(\omega_{it-1}^1) + \mathbb{1}(G_{it} = 1)h_1^+(\omega_{it-1}^0, \omega_{it-1}^1) \\ &\quad + \mathbb{1}(G_{it} = -1)h_1^-(\omega_{it-1}^0, \omega_{it-1}^1) + \epsilon_{it}^1, \\ \omega_{it}^0 &= \mathbb{1}(G_{it} = 0)\bar{h}_0(\omega_{it-1}^0) + \mathbb{1}(G_{it} = 1)h_0^+(\omega_{it-1}^0, \omega_{it-1}^1) \\ &\quad + \mathbb{1}(G_{it} = -1)h_0^-(\omega_{it-1}^0, \omega_{it-1}^1) + \epsilon_{it}^0. \end{aligned} \tag{C.3}$$

Compared to (3), equation (C.3) allows the firms to turn on and off the treatment across time and allows much more flexible transition dynamics when firms change their treatment status. As the identification of production function and realized productivity does not rely on the $G_{it} \neq 0$ periods, Theorem 3.1 still holds. The key difference is the definition of treatment effect and its identification.

Dynamic treatment effects are usually not identified under the context of volatile treatment. Instead, we focus on a particular category of treatment effects for firms that switch their treatment statuses at time g and maintain their new statuses for ℓ periods. Here we abuse the notation to use g to both denote the treatment cohort group and the group's

¹⁹See statistics on R&D participation reported by [Aw et al. \(2011\)](#); [Doraszelski and Jaumandreu \(2013\)](#); [Chen et al. \(2021\)](#).

²⁰See [Heckman and Navarro \(2007\)](#) for a formal definition of the general dynamic treatment effects.

initial treatment time. Formally, we define $ATT_{g,\ell}^+$ and $ATT_{g,\ell}^-$ as the ℓ -period persistent treatment effects for firms that, at time g , switch into or out of treatment, respectively:

$$\begin{aligned} ATT_{g,\ell}^+ &= \mathbb{E}[\omega_{ig+\ell}^1 - \omega_{ig+\ell}^0 | D_{ig-1} = 0, D_{ig} = \dots = D_{ig+\ell} = 1], \\ ATT_{g,\ell}^- &= \mathbb{E}[\omega_{ig+\ell}^1 - \omega_{ig+\ell}^0 | D_{ig-1} = 1, D_{ig} = \dots = D_{ig+\ell} = 0]. \end{aligned} \quad (C.4)$$

We first show that the 0-period ahead treatment effect is identified under the conditional parallel trend assumption for both negative and positive switchers.

Proposition C.3. *Under Assumption 4.1, the 0-period-ahead positive/negative switching ATT effects at time g are identified as $ATT_{g,0}^+ = \mathbb{E}[\omega_{ig} - \bar{h}_0(\omega_{ig-1}) | D_{ig-1} = 0, D_{ig} = 1]$, and $ATT_{g,0}^- = \mathbb{E}[\omega_{ig} - \bar{h}_1(\omega_{ig-1}) | D_{ig-1} = 1, D_{ig} = 0]$.*

Similar to the absorbing-treatment case, evaluating the ℓ -period-ahead ATT requires an additional structural assumption on the exogeneity of innovations.

Assumption C.2. *There is a cohort group g' such that all firms i' such that $i' \in g'$ are untreated within ℓ periods after period g' , i.e., $D_{ig'-1} = D_{ig'} = \dots = D_{ig'+\ell} = 0$. Moreover, the conditional distribution of $(\epsilon_{ig}^0, \dots, \epsilon_{ig+\ell}^0) | (i \in g, \omega_{ig-1}^0)$ is the same as the conditional distribution of $(\epsilon_{i'g'}^0, \dots, \epsilon_{i'g'+\ell}^0) | (i' \in g', \omega_{ig'-1}^0)$.*

Assumption C.2 generalizes Assumption 4.2 to the non-absorbing treatment case using firms that are not treated between g' and $g' + \ell$. As treatment is not absorbing, we further need to take conditional expectation under the lagged treatment D_{it-1} .

Proposition C.4. *Suppose Assumption 4.1 and C.2 hold. The $ATT_{g,\ell}^+$ is identified by the same expression as the $ATT_{g,\ell}$ in Proposition 4.2.*

The proof of Proposition C.4 is the same as the proof of Proposition 4.2 and is hence omitted here. Assumption C.2 has a similar restriction as Assumption 4.2. However, if firms are allowed to change the treatment status every period, then the g' -matching cohort is very hard to find: The ℓ -period untreated firms are likely to face a very high $\epsilon_{ig'}^0$, and hence these firms are not a good match for the g -cohort firms.

However, Assumption C.2 is likely to hold for treatments that must be maintained for several periods. For example, suppose that the treatment decision is whether to use a new technology and that the new technology is not available before the time g . If contractual obligations or the fixed cost of switching technology effectively guarantee that the firm's treatment status will persist for at least ℓ periods, we can use all firms at $g' = g - \ell$ as the match group.

D Additional Results for Alternative Models

D.1 Additional Moments for Restricted Productivity Processes

Our moment conditions in Proposition 3.1 impose no additional assumptions on the productivity evolution process (3). Although implementing moment conditions in Proposition 3.1 requires minimal structural assumptions, we require a relatively large sample of two-year consecutively untreated and treated observations as in Assumption 3.3. Such data requirements can be satisfied when the panel satisfies an absorbing treatment type design. However, if the treatment variable is volatile over time, we may need to discard a substantial fraction of the firms to implement (8) and (9), which leads to inefficient use of data. We now consider several alternative assumptions on the evolution process that allow us to derive more flexible moment conditions and make use of firms with volatile treatment status.

D.1.1 Independent Evolution Process

Let's consider the case where the two potential productivity processes evolve independently as in Example 3. In this case, we may substitute the Markov process back several periods to form additional moment conditions. Even for a firm that changes its treatment status every period, we know the treatment statuses every two periods must coincide. To form moment conditions for an independent productivity process, we impose the following assumption:

Assumption D.1. For $d = 0, 1$, the Markov process ω_{it}^d satisfies

$$\omega_{it}^d = \bar{h}_d^{(s)}(\omega_{it-s}^d) + r(\epsilon_{it}^d, \dots, \epsilon_{it-s+1}^d),$$

where $\bar{h}_d^{(s)}$ is an s -period transition function and $r(\cdot)$ is linear in all arguments.

Assumption D.1 is satisfied for the AR(1) process. The linearity of $r(\cdot)$ ensures that we can generalize moment conditions (8) and (9) to an s -period lagged evolution process. Note that we have to rule out the evolution process h_d^+ and h_d^- for the transition periods.

Corollary D.1. Suppose Assumption D.1 holds and the productivity process satisfies Example 3, then the following two moment conditions hold:

$$\mathbb{E}[\omega_{it}(\beta) - \bar{h}_0^{(s)}(\omega_{it-s}(\beta)) | \mathcal{Z}_{it-s+1}, D_{it} = D_{it-s} = 0] = 0, \quad (\text{D.1})$$

$$\mathbb{E}[\omega_{it}(\beta) - \bar{h}_1^{(s)}(\omega_{it-s}(\beta)) | \mathcal{Z}_{it-s+1}, D_{it} = D_{it-s} = 1] = 0. \quad (\text{D.2})$$

Moment conditions (D.1) and (D.2) allow us to use a larger fraction of firms in the dataset. However, we recommend combining moment conditions (D.1) and (D.2) with (8) and (9) to estimate the production functions unless Assumption 3.3 fails. It's unfortunate that we cannot show nonparametric identification of production function with moments (D.1) and (D.2) alone: The error term ϵ_{it-s} is correlated with k_{it} and l_{it} for all $s \geq 1$, and thus they are not in the instrument set $\mathcal{Z}_{it-s+1}$. Therefore, we cannot apply the GNR trick to differentiate both sides of (D.1) to identify the production function.

One may argue that k_{it-s+1} and l_{it-s+1} can serve as instruments for k_{it} and l_{it} . However, without solving the firms' dynamic optimization problem, we cannot establish the functional relationship between (k_{it}, l_{it}) and (k_{it-s+1}, l_{it-s+1}) , and we cannot prove the nonparametric identification of production functions. However, when the production function is Cobb-Douglas, the log-linear form of the production function along with the valid instrument k_{it-s+1} and l_{it-s+1} allow us to identify the production function parameters and the evolution process.

D.1.2 Divergent Productivity Processes

Now we consider the productivity process in Example 2. As only the observed productivity matters for the evolution process, we can further derive the moment conditions at the transition periods.

Corollary D.2. *Suppose Assumptions 2.1-3.3 hold and the productivity evolution process satisfies Example 2. Then the moment conditions (4) (and respectively (6)), (8), (9) and*

$$\mathbb{E}[\omega_{it}(\beta) - h_1^+(\omega_{it-1}(\beta)) | \mathcal{Z}_{it}, D_{it} = 1, D_{it-1} = 0] = 0, \quad (\text{D.3})$$

identify the production functions, and the evolution processes $\bar{h}_d$ and h_1^+ nonparametrically up to a constant difference.

The additional moment conditions in Corollary D.2 are useful when the panel is short or when we only observe one period after the treatment status changes. Corollary D.2 requires the transition period to be treated separately from the consistent treatment status period. Moment condition (D.3) is imposed to identify the positive transition process h^+ .

D.2 Identifying CATT with an Alternative Assumption

Assumption D.2. *The Markov process ω_{it}^0 satisfies*

$$\omega_{it}^0 = \bar{h}_0^{(s)}(\omega_{it-s}^0) + r(\epsilon_{it}^0, \dots, \epsilon_{it-s+1}^0),$$

where $\bar{h}_0^{(s)}$ is an s -period transition function and $r(\cdot)$ is linear in all its arguments. Moreover, the $E[\epsilon_{it-s+\ell}^0 | \omega_{it-s}^0] = 0$ for all $\ell \geq 0$.

Proposition D.1. Under Assumption 3.2, 4.1, and D.2, the ℓ -period-ahead CATT is identified as $CATT_\ell(\omega) = \mathbb{E}[\omega_{ie_i+\ell} - \bar{h}_0^{(\ell)}(\omega_{ie_i-1}) | \omega_{ie_i-1} = \omega]$.

Proof. Note that conditional on $e_i = t$,

$$\begin{aligned} (CATT_\ell(\omega) | e_i = t) &=_{(i)} \mathbb{E}[\omega_{it+\ell} - \bar{h}_0^{(\ell)}(\omega_{it-1}^0) - r(\epsilon_{it}^0, \dots, \epsilon_{it-s+1}^0) | e_i = t, \omega_{ie_i-1} = \omega] \\ &=_{(ii)} \mathbb{E}[\omega_{it+\ell} - \bar{h}_0^{(\ell)}(\omega_{it-1}) | e_i = t, \omega_{ie_i-1} = \omega], \end{aligned}$$

where (i) by substituting the ω_{it}^0 with the evolution process in Assumption D.2. Note that the treatment is absorbing, so

$$E[r(\epsilon_{it}^0, \dots, \epsilon_{it-s+1}^0) | e_i = t, \omega_{ie_i-1} = \omega] = E[r(\epsilon_{it}^0, \dots, \epsilon_{it-s+1}^0) | D_{it} = 1, \omega_{ie_i-1} = \omega].$$

As a result, (ii) follows by Assumptions 3.2 and linearity of $r(\cdot)$. The result in the proposition follows by further taking expectations with respect to e_i . $\square$

Assumption D.2 is satisfied for an AR(1) productivity process, but generally fails when non-linearity appears in the transition function $\bar{h}_0$. Therefore, Assumption D.2 can be restrictive.

D.3 Counterfactual Treatment Effect

Treatment effect objects such as ATT and ATE are useful when we take a retrospective evaluation of the treatment or policy effect. However, in many settings, policymakers are deciding whether to apply the same treatment policy to a counterfactual group of firms based on the knowledge from the currently available data.

In this section, we consider a program that rolls out in several phases and the treatment status is absorbing. We start with an initial full set of firms (denoted by S) that are not treated. At time t_0 , a subset of firms become treated (denoted by S^{tr}) whereas the rest of the firms remain untreated (denoted by S^{ut}). Untreated firms cannot change their treatment status unless a new phase of the program begins. A policymaker stands at period $t_0 + s$ and has access to firm-level data up to time $t_0 + s - 1$ and needs to decide whether to start a new phase of the program. There are many empirical examples where the treatment program rolls out in several phases: For example, the State-Owned Enterprise reform in China²¹ first took place in the 18 experimental cities and rolled out to the

²¹This is known as the privatization process of the state-owned enterprises.

rest of the country in several phases.

The policymaker is interested in the treatment effects on the untreated group $\mathcal{S}^{ut}$, whereas the treatment effects identified in previous sections are evaluated using the whole sample $\mathcal{S}$. These two quantities in general do not coincide even when the policy is a fully randomized controlled experiment. This is because the treatment effect objects at time t_0 depend on the distribution of potential outcome $\omega_{it_0-1}^1$. Although a fully randomized treatment ensures that $F(\omega_{it_0-1}^1 | i \in \mathcal{S}^{tr}) = F(\omega_{it_0-1}^1)$, the s -period ahead distribution of potential outcome $\omega_{it_0+s-1}^1$ will not be the same as the $t_0 - 1$ period potential outcome distribution, i.e., $F(\omega_{it_0+s-1}^1 | i \in \mathcal{S}^{tr}) \neq F(\omega_{it_0-1}^1)$, unless the productivity distribution is stationary.

We, therefore, seek to characterize the counterfactual treatment effect objects that allow the policymaker to evaluate the value of extending the program to the rest of the firms at time $t + s$. In general, without imposing further structural assumptions other than Assumptions 2.2-A.3, it is almost impossible to identify the counterfactual treatment effect objects: the target treatment effect is defined as the difference $\omega_{it_0+s}^1 - \omega_{it_0+s}^0$, but for the $\mathcal{S}^{ut}$ firms, we have at best the knowledge of $\omega_{it_0+s-1}^0$ but not $\omega_{it_0+s-1}^1$. We therefore consider several additional structural assumptions that allow us to evaluate the counterfactual treatment effects defined in the following:

$$ATE_{s,\ell}^{count} \equiv E[\omega_{it_0+s+\ell}^1 - \omega_{it_0+s+\ell}^0 | i \in \mathcal{G}], \quad (\text{D.4})$$

which is the ℓ -period ahead counterfactual treatment effect for group $\mathcal{G} \subseteq \mathcal{S}^{ut}$ firms if the treatment takes place at time $t_0 + s$.

D.3.1 Divergent Productivity Processes

Recall that the difficulty of characterizing the counterfactual treatment effect comes from the lack of knowledge of $\omega_{it_0+s-1}^1$. However, the divergent productivity process in Example 2 implies the coincidence of two potential outcomes before treatment status changes: $\omega_{it_0+s-1}^0 = \omega_{it_0+s-1}^1$ for all untreated firms $i \in \mathcal{S}^{ut}$ and $s \leq 0$. Therefore, we can characterize the counterfactual treatment effect.

Proposition D.2. *Let the productivity evolution process satisfy Example 2. Moreover, suppose the conditional parallel trend assumption 4.1 holds. For a subset $\mathcal{G} \subseteq \mathcal{S}^{ut}$ of not-yet treated firms at time $t_0 + s$ that are assigned to take treatment at $t_0 + s$, the instantaneous counterfactual treatment effect is identified as*

$$ATE_{s,0}^{count} = E[h_1^+(\omega_{it_0+s-1}) - \bar{h}_0(\omega_{it_0+s-1}) | i \in \mathcal{G}],$$

where h_1^+ is identified from Corollary D.2.

Proof. By the divergent productivity process assumption, $\omega_{it_0+s}^1 = h_1^+(\omega_{it_0+s-1}) + \epsilon_{it_0+s}^1$ and $\omega_{it_0+s}^0 = \bar{h}_0(\omega_{it_0+s-1}) + \epsilon_{it_0+s}^0$. The result follows by the conditional mean zero condition: $E[\epsilon_{it_0+s}^d | D_{it_0+s}] = 0$ for $d \in \{0, 1\}$. $\square$

For ℓ -period ahead counterfactual treatment effect, we need additional structural assumptions on the productivity process innovations so that we can simulate the productivity process several periods ahead.

Assumption D.3. (i). The innovations satisfy $\epsilon_{it}^d \sim_{i.i.d} G_\epsilon^d(\cdot)$ for $d \in \{0, 1\}$, where the i.i.d is over both firm index i and time index t . (ii). No selection in pre-treatment innovations: $\epsilon_{it}^0 \sim_{i.i.d} G_\epsilon^0(\cdot)$ for $t < t_0$. (iii) No selection in already-treated group innovations: $\epsilon_{it}^1 | i \in \mathcal{S}^{tr} \sim_{i.i.d} G_\epsilon^1(\cdot)$ for $t_0 \leq t < t_0 + s$.

Assumption D.3 is similar to Assumption 4.3 except that we also require that the distribution $G_\epsilon^1(\cdot)$ is identified from the already treated firms $\mathcal{S}^{tr}$. This is because, for the factual treatment, we can observe the ω_{it+s}^1 and ϵ_{it+s}^1 once the firms are treated. However, for the counterfactual treatment effects, we need to simulate both the treated and untreated future productivity.

Proposition D.3. Under Assumption 4.1, 4.2, and D.3, $G_\epsilon^0, G_\epsilon^1$ are identified, and the ℓ -period-ahead counterfactual treatment effect at period $t_0 + s$ is identified as

$$\begin{aligned} ATE_{s,\ell}^{count} &= \mathbb{E}_{(G_\epsilon^1)^\ell} [\bar{h}_1^{(\ell-1)}(h_1^+(\omega_{it_0+s-1}, \epsilon_{it_0+s}^1), \epsilon_{it_0+s+1}^1, \dots, \epsilon_{it_0+s+\ell}^1) | i \in \mathcal{G}] \\ &\quad - \mathbb{E}_{(G_\epsilon^0)^\ell} [\bar{h}_0^{(\ell)}(\omega_{it_0+s-1}, \epsilon_{is}^0, \dots, \epsilon_{is+\ell}^0) | i \in \mathcal{G}], \end{aligned}$$

where the expectation on $(G_\epsilon^d)^\ell$ is taken over the joint distribution of $(\epsilon_{it_0+s}^d, \dots, \epsilon_{it_0+s+\ell}^d)$.

Remark D.1. The characterization of the counterfactual treatment effect also highlights another reason in favor of the potential productivity process over the endogenous productivity method (Doraszelski and Jaumandreu, 2013). Recall that Doraszelski and Jaumandreu (2013) do not model the transition period and implicitly assume that $h_1^+ = \bar{h}_1$ in the identifying moment condition. Although imposing $h_1^+ = \bar{h}_1$ may not lead to a large bias in the production function estimates when the panel is long, it does lead to a bias in the counterfactual treatment effect, especially the instantaneous treatment effect $ATE_{s,0}^{count}$.

D.3.2 Stationary Conditional Potential Outcome Moments

In more general models, we do not have information on the ω_{it+s-1}^1 for the not-yet-treated group $\mathcal{S}^{ut}$. We now investigate conditions where we can transfer the knowledge of the

factual treatment effect to the counterfactual treatment effect. In particular, we want the stationary conditional distribution of potential outcomes:

Assumption D.4. *The distribution of $\omega_{it_0+s-1}^1 | (\omega_{it_0-1}^0 = w, i \in \mathcal{S}^{tr})$ is the same as the distribution of $\omega_{it_0+s-1}^1 | (\omega_{it_0-1}^0 = w, i \in \mathcal{G})$.*

Assumption D.4 is the high-level assumption on potential productivity distribution. There are two constraints embedded in D.4: (i) no selection on the potential outcome. This is reflected in the requirement that we condition on the different firm groups $\mathcal{S}^{tr}$ and $\mathcal{S}^{ut}$; (ii) the conditional distribution of ω_{it}^1 is stationary at time $t_0 - 1$ and $t_0 + s - 1$.

Proposition D.4. *Suppose Assumptions D.3 and D.4 hold. The counterfactual treatment effect is identified as*

$$ATE_{s,\ell}^{count} = \mathbb{E} \left\{ \mathbb{E}[\omega_{i't_0+\ell} | i' \in \mathcal{S}^{tr}, \omega_{i't_0-1} = \omega_{it_0+s-1}] \middle| i \in \mathcal{G} \right\} - \mathbb{E}_{(G_\epsilon^0)^\ell} [\bar{h}_0^{(\ell)}(\omega_{it_0+s-1}, \epsilon_{is}^0, \dots, \epsilon_{is+\ell}^0) | i \in \mathcal{G}].$$

Proof. We first note that

$$\begin{aligned} & \mathbb{E}[\omega_{i't_0+\ell} | i' \in \mathcal{S}^{tr}, \omega_{i't_0-1} = \omega_{it_0+s-1}] \\ &= \mathbb{E}_{(G_\epsilon^1)^\ell, \omega_{i't_0-1}^1} [\bar{h}_1^{(\ell-1)}(h_1^+(\omega_{i't_0-1}^1, \omega_{i't_0-1}^0, \epsilon_{it_0}^1), \epsilon_{i't_0+1}^1, \dots, \epsilon_{i't_0+\ell}^1) | i' \in \mathcal{S}^{tr}, \omega_{i't_0-1} = \omega_{it_0+s-1}] \\ &= (*) \mathbb{E}_{(G_\epsilon^1)^\ell, \omega_{it_0+s-1}^1} [\bar{h}_1^{(\ell-1)}(h_1^+(\omega_{it_0+s-1}^1, \omega_{it_0+s-1}^0, \epsilon_{it_0+s}^1), \epsilon_{it_0+s+1}^1, \dots, \epsilon_{it_0+s+\ell}^1) | i \in \mathcal{G}, \omega_{it_0+s-1}] \\ &= \mathbb{E}[\omega_{it_0+s+\ell}^1 | i \in \mathcal{G}, \omega_{it_0+s-1}], \end{aligned}$$

where we use Assumptions D.3 and D.4 in the (*) step.

Then the counterfactual treatment effect is identified as

$$\begin{aligned} ATE_{s,\ell}^{count} &= \mathbb{E} \left\{ \mathbb{E}[\omega_{it_0+s+\ell}^1 | i \in \mathcal{G}, \omega_{it_0+s-1}] - \mathbb{E}[\omega_{it_0+s+\ell}^0 | i \in \mathcal{G}, \omega_{it_0+s-1}] \middle| i \in \mathcal{G} \right\} \\ &= \mathbb{E} \left\{ \mathbb{E}[\omega_{i't_0+\ell} | i' \in \mathcal{S}^{tr}, \omega_{i't_0-1} = \omega_{it_0+s-1}] \middle| i \in \mathcal{G} \right\} \\ &\quad - \mathbb{E}_{(G_\epsilon^0)^\ell} [\bar{h}_0^{(\ell)}(\omega_{it_0+s-1}, \epsilon_{is}^0, \dots, \epsilon_{is+\ell}^0) | i \in \mathcal{G}]. \end{aligned}$$

The result follows. □

The identified counterfactual treatment effect in Proposition D.4 uses two different approaches to impute the unrealized future potential productivities. For the treated future potential productivity, we use the stationary distribution assumption and use the already treated firms to impute the $\omega_{it_0+s+\ell}$. In particular, we match each not-yet-treated firm at

time $t_0 + s - 1$ with an already-treated firm at time $t_0 - 1$ with the same realized productivity. For the untreated potential future productivity, we simulate the productivity process into the future.

D.4 The Misspecification in Existing Methods

The Ex-post Regression The problem with this approach is that the realized productivity only satisfies the moment condition (5) when $\omega_{it}^1 - \bar{h}_1(\omega_{it}^1) = \omega_{it}^0 - \bar{h}_0(\omega_{it}^0)$ and $h_1^+(\omega_{it}^0, \omega_{it}^1) = \bar{h}_0(\omega_{it}^0)$. If the treatment has a trend effect or a contemporaneous effect, then the estimated production function will generally be biased.

To demonstrate the source of the specification bias, we first specialize the moment condition to this example in the pre-treatment periods $t \leq T_0$. Equation (5) becomes

$$\mathbb{E}[\omega_{it}^0(\beta) - \bar{h}_0(\omega_{it-1}^0(\beta)) | \mathcal{Z}'_{it}] = 0 \quad \forall t \leq T_0.$$

where $\mathcal{Z}'_{it} = \{k_{it}, l_{it}, k_{it-1}, l_{it-1}, m_{it-1}\}$. By Proposition 3.1, this moment condition identifies β and $\bar{h}_0$. Substituting β and $\bar{h}_0$ for h into (5) for $t = T_0 + 1$ and $t > T_0 + 1$, we obtain

$$0 = \mathbb{E}[\omega_{iT_0+1}^1 - \bar{h}_0(\omega_{iT_0}^0) | \mathcal{Z}'_{iT_0+1}] Pr(D_{iT_0+1} = 1) + \mathbb{E}[\omega_{iT_0+1}^0 - \bar{h}_0(\omega_{iT_0}^0) | \mathcal{Z}'_{iT_0+1}] Pr(D_{iT_0+1} = 0), \quad (\text{D.5})$$

$$0 = \mathbb{E}[\omega_{it}^1 - \bar{h}_0(\omega_{it-1}^1) | \mathcal{Z}'_{it}] Pr(D_{it} = 1) + \mathbb{E}[\omega_{it}^0 - \bar{h}_0(\omega_{it-1}^0) | \mathcal{Z}'_{it}] Pr(D_{it} = 0), \quad \forall t > T_0 + 1. \quad (\text{D.6})$$

In deriving these equations, we have used the fact that $\omega_{it}(\beta) = \omega_{it}^{D_{it}}$ because β is the true parameter vector and the assumption that treatment status is exogenously assigned. The second additive terms in equations (D.5) and (D.6) are zero by construction. However, if the policy has any effect on the expected evolution of productivity, the first terms will generally not be zero. In particular, (D.5) will be violated unless $h_1^+(\omega_{it}^0, \omega_{it}^1) = \bar{h}_0(\omega_{it}^0)$, whereas (D.6) fails whenever $\bar{h}_1 \neq \bar{h}_0$.

The Endogenous Productivity Method Indeed, by defining $\bar{h}_d(\cdot) = \tilde{h}(\cdot, d)$ for $d = 0, 1$, we can show that moment condition (5) can be transformed into

$$\begin{aligned} & \mathbb{E}[\omega_{it}^0(\beta) - \bar{h}_0(\omega_{it-1}^0(\beta)) | \mathcal{Z}_{it}] = 0 \quad \forall t \leq T_0, \quad \text{and} \\ & \mathbb{E}[\omega_{it}^0(\beta) - \bar{h}_0(\omega_{it-1}^0(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 0] Pr(D_{it} = D_{it-1} = 0) \\ & + \mathbb{E}[\omega_{it}^1(\beta) - \bar{h}_1(\omega_{it-1}^1(\beta)) | \mathcal{Z}_{it}, D_{it} = D_{it-1} = 1] Pr(D_{it} = D_{it-1} = 1) = 0 \quad \forall t > T_0 + 1, \end{aligned} \quad (\text{D.7})$$

and the moment condition in the first period of treatment ($T_0 + 1$) becomes

$$\underbrace{\mathbb{E}[\omega_{iT_0+1}^0(\beta) - \bar{h}_0(\omega_{iT_0}^0(\beta)) | \mathcal{Z}'_{iT_0+1}, D_{iT_0+1} = D_{iT_0} = 0]}_{\text{Part A}} Pr(D_{iT_0+1} = D_{iT_0} = 0) + \underbrace{\mathbb{E}[\omega_{iT_0+1}^1(\beta) - \bar{h}_1(\omega_{iT_0}^0(\beta)) | \mathcal{Z}'_{iT_0+1}, D_{iT_0+1} = 1, D_{iT_0} = 0]}_{\text{Part B}} Pr(D_{iT_0+1} = 1, D_{iT_0} = 0) = 0. \quad (\text{D.8})$$

The moment conditions in equation (D.7) are correctly specified. In particular, by Proposition 3.1, β and $\bar{h}_0$ are identified from the $t \leq T_0$ moment equality (D.7), and $\bar{h}_1$ is identified from the $t \geq T_0 + 2$ moment equality (D.7).

However, the moment condition for the initial treatment period (D.8) is misspecified. Substituting the value of $\bar{h}_0$ identified from (D.7), Part A in (D.8) equals zero. However, Part B is generally nonzero under the general evolution process (3). The transition function in the first treatment period should be $h_1^+(\omega_{it-1}^0, \omega_{it-1}^1)$, whereas in Part B of (D.8), the transition function is $\bar{h}_1(\omega_{it-1}^0)$. If either the transition function has a contemporaneous effect so that $\omega_{it-1}^0 \neq \omega_{it-1}^1$ or the expected evolution in the initial period of treatment is different so that $h_1^+(\omega_{it-1}^0, \omega_{it-1}^1) \neq \bar{h}_1(\omega_{it-1}^1)$, then the moment condition will not be satisfied.

E Data Appendix

E.1 Construction of Other Variables

We construct the main variables for production function estimation as follows.

Materials: Costs of goods sold plus selling, general and administrative expenses minus labor costs. Labor costs are measured using the payroll payable, and deflated using the industry-year level input price index.

Capital: Fixed assets including property, plant, and equipment (PP&E) deflated by province-year level investment price index.

Labor: Total number of registered working employees reported in the annual report.

Value Added: Operational revenue minus materials, deflated by province-year level output price index.

Annual Sales: Total operational revenue, deflated province-year level output price index.

All the price indices are extracted from China's Statistical Yearbook. The summary statistics of these variables are displayed in the Table E.1.

Table E.1: Summary Statistics of Main Production Variables

Variable	Mean	SD	P5	P25	P50	P75	P95
m	21.084	1.276	19.215	20.184	20.984	21.828	23.394
l	7.667	1.068	6.016	6.921	7.609	8.341	9.484
k	19.914	1.280	17.964	19.083	19.835	20.655	22.151
y	19.047	1.250	17.232	18.204	18.928	19.793	21.290
$\ln(sale)$	21.149	1.236	19.357	20.284	21.035	21.870	23.411
$Digit$	0.082	0.275	0.000	0.000	0.000	0.000	1.000

The industrial classification is based on the two-digit China's National Industrial Classification. We choose the manufacturing industries and perform the estimation by 2-digit industry. We drop some industries that contain too few observations to conduct meaningful analysis or contain too few treated observations. The final sample of industries and number of observations for treated and control groups are listed in Table E.2.

Table E.2: Number of Treated and Untreated Observations for Different Industries

Industries	Untreated	Treated	Total
Healthcare Manufacturing	1,321	53	1,374
Metal Processing	1,149	58	1,207
Food & Beverage	628	59	687
Electronics Manufacturing	1,085	63	1,148
Chemical Synthesis	1,594	75	1,669
Equipment Manufacturing	3,139	493	3,632
Total	8,916	801	9,717

Note: The Electronics Manufacturing industry encompasses the production of various electronic equipment, including the manufacturing of other electronic equipment, daily-use electronic appliances, and electronic components. The Equipment Manufacturing industry involves the production of specialized equipment, transportation equipment, instrumentation, cultural and office machinery, general machinery, and electrical machinery and equipment. The Healthcare Manufacturing industry specializes in the production of pharmaceuticals and biotechnology products. The Food & Beverage industry focuses on food manufacturing, food processing, and beverage manufacturing. The Metal Processing industry encompasses various activities, including nonferrous metal smelting and rolling, metal product manufacturing, non-metallic mineral product manufacturing, and ferrous metal smelting and rolling. The Chemical Synthesis industry includes the manufacturing of chemical raw materials and chemical products, chemical fiber, plastics, petroleum processing and coking, and rubber products.

E.2 Defining Production Digitalization

The text analysis of annual reports contains two main steps: keyword searching and refining.

Step 1: Keywords Searching To capture state-of-the-art digital technologies involved in production digitalization, we choose the following keywords (Chinese bopomofo in brackets): *digitalization* (Shu Zi Hua), *smartness* (Zhi Neng), *intelligence* (Zhi Hui), *Internet of Things* (Wu Lian Wang or IoT), *industrial internet* (Gong Ye Wu Lian Wang), *big data* (Da Shu Ju), *cloud computing* (Yun Ji Suan), *industrial cloud* (Gong Ye Yun), *platform* (Ping Tai), *SaaS*, *C2M* and *various management information systems* (such as PDM, ERP, SRM, CRM, MES, SCADA, PLM and their Corresponding Chinese names), “*Smart*” (Zhi Neng), “*Intelligent*” (Zhi Hui), and “*Platform*” (Ping Tai). These keywords appear in annual reports in various forms, such as “*Smart Manufacturing*” (Zhi Neng Zhi Zao), “*Smart Factory*” (Zhi Hui Gong Chang), “*Smart Production*” (Zhi Neng Zhi Zao), “*Smart Firms*” (Zhi Hui Xing Qi Ye), “*Cloud Platform*” (Yun Ping Tai), and “*Digital Purchasing Platform*” (Dian Zi Cai Gou Ping Tai), etc. To avoid missing useful information on digitalization, we only use the stem words “*Zhi Neng*”, “*Zhi Hui*”, and “*Platform*” to identify digitalization-related texts.

Step 2: Manual Reading and Refining By manual reading of the annual reports, any paragraphs on digitalization that are related to production, manufacturing and equipment or workshop upgrade are classified as production digitalization. However, we notice that in some annual reports, firms may describe the development of digitalization in their own sector or China’s national strategy, which is not related to the firm’s own implementation of digitalization. In our construction of the digitalization indicator, we exclude such scenarios by manually identifying them and excluding them from the firm’s own digitalization strategy. We list three examples below:²²

- **Example 1** (Stock ID: 000008, Year: 2018) “Driven by the trend of technological progress, rail transit operation and maintenance equipment have been upgraded from informatization to digitalization characterized by intelligence, data, internet and deep learning. Traditional equipment is upgrading to intelligent equipment; the operation and maintenance system is upgrading from single-product intelligence to the unmanned maintenance factory. It is the right time for data-oriented development of equipment in rail transit industry.”

²²The English texts are translated from Chinese texts in firms’ annual reports.

- **Example 2** (Stock ID: 300161, Year: 2017): “*Made in China 2025* puts forward the strategic goal of achieving manufacturing power through ‘three steps’. Centering on the top-level design of *Made in China 2025*, relevant supporting policies have been issued successively, and *intelligent manufacturing pilot demonstration projects have been accelerated, with obvious demonstration effect*. With the further deepening of transformation, China’s manufacturing industry will be further enhanced in digitalization, networking and intelligence.”
- **Example 3** (Stock ID: 000020, Year: 2012) “...domestic and international economic environment is complex, with difficult concerns and positive factors coexisting. On the one hand, the ability of technological innovation is insufficient. *In the new wave of industrial revolution which centers on global digital and intelligent manufacturing, the gap between domestic enterprises and developed countries in Europe and the United States in the field of high-end technology is facing the risk of being widened again, and enterprises will bear the pain of structural adjustment in the process of industrial upgrading ...*”

The quoted paragraph in *Example 1* talks about the digitalization development in its own industry, but not the firm’s own digitalization. In *Example 2*, the paragraph is a description of China’s national strategy for digitalization. *Example 3* mentions the global environment of digitalization, but not the firm’s own strategy.

Examples of Identified Production Digitalization To be concrete, we provide some examples of texts that are identified as production digitalization after performing the text analysis:

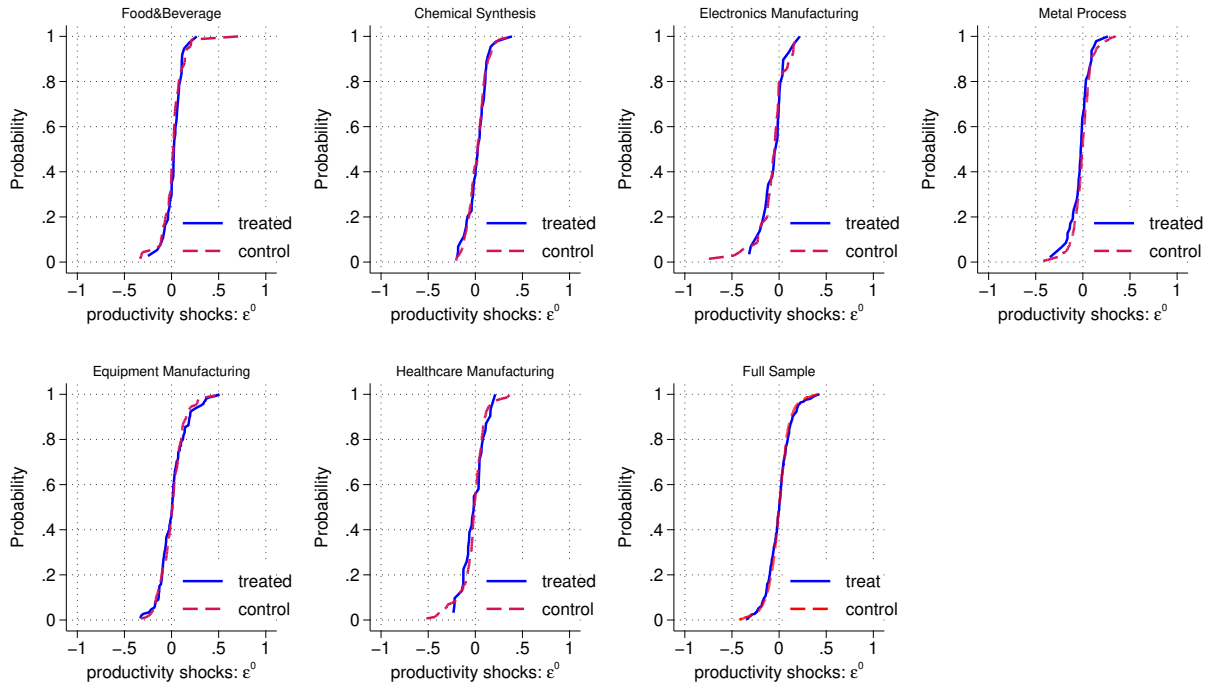
- **Example 1** (Stock ID: 002085, Year: 2018): “Our company has intensified the transformation and upgrading efforts, established intelligent factories with robots as the core, improved the automation level of manufacturing industries, and improved the core competitiveness. By building a digital platform in the whole field of digital research and development, digital technology and digital manufacturing of Wanfeng, our company optimized and standardized the operating system, realized the product life cycle management, and provided data support for company information construction and intelligent manufacturing of intelligent factory.”
- **Example 2** (Stock ID: 002920, Year: 2018): “.....The company has built a digital intelligent factory in an all-round way and established industry-leading highly automated and information-based production lines. Now digital intelligent factories

and intelligent storage systems have been put into use successively. The construction project of integrated industrialization of automobile electronics and mobile Internet technology has officially laid the foundation and is under construction.”

E.3 Suggestive Evidence of Identification Assumptions for the Empirical Study

Figure E.1 presents the empirical cumulative distribution function (CDF) of ϵ_{it}^0 for treated and control units during the three years preceding the initial year of adopting production digitalization for each industry. Table E.3 reports the results of the Kolmogorov-Smirnov test for the CDFs of ϵ_{it}^0 between treated and control units. The null hypothesis of the K-S test is that the two samples are drawn from the same distribution. The p-values in parentheses indicate that we cannot reject the null hypothesis for all industries, suggesting that the distributions of ϵ_{it}^0 for treated and control units are similar before the initial treatment adoption. This provides evidence for Assumption 4.2.

Figure E.1: Empirical Cumulative Distribution of ϵ_{it}^0



Note: The blue solid line (*resp.* red dashed line) represents the CDF of ϵ_{it}^0 for treated units (*resp.* control units) during three years preceding the initial treatment adoption.

Table E.3: Results of Kolmogorov-Smirnov equality-of-distributions test

Industry	Control<Treated	Treated < Control	Combined K-S
Food & Beverage	0.156 (0.310)	-0.076 (0.755)	0.156 (0.602)
Chemical Synthesis	0.101 (0.493)	-0.049 (0.846)	0.101 (0.871)
Electronics Manufacturing	0.135 (0.475)	-0.122 (0.545)	0.135 (0.851)
Metal Processing	0.030 (0.935)	-0.148 (0.187)	0.148 (0.372)
Equipment Manufacturing	0.072 (0.448)	-0.099 (0.221)	0.099 (0.438)
Healthcare Manufacturing	0.132 (0.410)	-0.135 (0.391)	0.135 (0.736)
All	0.044 (0.396)	-0.054 (0.249)	0.054 (0.491)

Note: The column “Control<Treated” reports the maximum difference where treated firms have higher values for ϵ_{it}^0 , the column “Treated<Control” represents the maximum difference where the control firms have higher values for ϵ_{it}^0 , and the column “Combined K-S” shows the usual two-sided K-S statistic for the absolute maximum difference. The p-values for the test statistics are in parentheses.

E.4 Supplementary Empirical Results

Estimates of the Translog Production Functions Table E.4 displays the estimates of the translog production functions for the six industries in the sample. Note that there are large differences in the production function coefficient estimates, indicating the necessity of estimating the production function separately for each industry. Moreover, almost all the coefficient estimates are statistically significant, confirming the plausibility of using the translog specification to allow the output-input elasticities to depend on the input levels.

Table E.4: Estimates of Translog Production Functions

Industries	β_l	β_k	β_{ll}	β_{kk}	β_{lk}	β_t
Food & Beverage	0.981 (0.006)	-3.549 (0.006)	-0.099 (0.007)	0.089 (0.006)	0.050 (0.010)	0.025 (0.020)
Chemical Synthesis	1.070 (0.005)	-1.973 (0.005)	0.107 (0.007)	0.079 (0.005)	-0.113 (0.009)	0.035 (0.009)
Electronics Manufacturing	1.332 (0.002)	-2.452 (0.003)	0.123 (0.004)	0.089 (0.002)	-0.137 (0.004)	0.052 (0.011)
Metal Processing	0.140 (0.002)	-0.735 (0.004)	0.148 (0.003)	0.034 (0.003)	-0.080 (0.004)	0.034 (0.008)
Equipment Manufacturing	1.629 (0.001)	-3.370 (0.002)	0.068 (0.002)	0.110 (0.001)	-0.110 (0.002)	0.043 (0.005)
Healthcare Manufacturing	2.334 (0.004)	-3.841 (0.005)	0.184 (0.008)	0.148 (0.004)	-0.234 (0.008)	0.055 (0.008)

Note: The standard errors in the parentheses are obtained by bootstrapping 500 times.

Regression Results using Other Methods In Table E.5, we present the regression results using alternative methods to estimate the effects of production digitalization on productivity. Panel A reports the results for the electronics manufacturing industry, and Panel B reports the results for the full sample. “Method I” is the ex-post regression method with exogenous productivity process (19), “Method II” is the ex-post regression method with endogenous productivity process (20) using the full sample, and “Method III” is the ex-post regression method with endogenous productivity process (20) excluding the transition period. For each method, the first column has no lagged productivity, the second column includes lagged first-order productivity, and the third column includes lagged first-, second- and third-order productivity. We control firm and year fixed effects in all regressions.

The results presented in Panel A have been summarized and discussed in the main text. Panel B reports additional regression results that incorporate lagged productivity terms. Consistent with our earlier findings, the full-sample regression in Panel B shows no significant productivity effects of production digitalization when lagged productivity is included as a control. Recall that our method estimates a slightly positive but statistically insignificant treatment effect for the entire sample. In this context, whether the treatment indicator is included directly in the productivity process has limited influence on the production function estimation, which explains the similarity between the results reported in columns (3) and (6). Furthermore, when we exclude the transition period from the estimation of the production function, the coefficient estimate on Digit_{it} changes from -0.0048 in column (6) to -0.0009 in column (9), which closely aligns with the average

treatment effect estimated by our method. This pattern suggests that the transition period plays an important role in shaping the measured productivity effects of production digitalization.

Table E.5: Productivity Effects Estimation Results Ex-post Regression Methods: Full Sample and Sample of Electronics Manufacturing Industry

Panel A: Industry of Electronics Manufacturing

Variables	Method I			Method II			Method III		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Digit_{it}	0.304*** (0.0683)	0.0318 (0.0218)	0.0255 (0.0217)	0.303*** (0.0682)	0.0317 (0.0218)	0.0254 (0.0217)	0.266*** (0.0609)	0.0320 (0.0232)	0.0284 (0.0229)
ω_{it-1}		0.951*** (0.0101)	-379.200*** (88.1300)		0.951*** (0.0101)	-383.200*** (88.1600)		0.923*** (0.0122)	-541.700*** (112.5000)
ω_{it-1}^2			10.690*** (2.4810)			10.810*** (2.4830)			12.120*** (2.5280)
ω_{it-1}^3			-0.100*** (0.0233)			-0.101*** (0.0233)			-0.090*** (0.0189)
N	992			992			968		

Panel B: Full Sample

Variables	Method I			Method II			Method III		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Digit_{it}	0.076*** (0.0177)	-0.0048 (0.0068)	-0.0049 (0.0068)	0.077*** (0.0177)	-0.0047 (0.0068)	-0.0048 (0.0068)	0.081*** (0.0175)	-0.0001 (0.0073)	-0.0009 (0.0073)
ω_{it-1}		0.912*** (0.0041)	0.703*** (0.1099)		0.912*** (0.0041)	0.700*** (0.1101)		0.899*** (0.0046)	-167.200*** (17.080)
ω_{it-1}^2			0.007** (0.0034)			0.007** (0.0034)			3.807*** (0.3902)
ω_{it-1}^3			-0.00008** (0.00003)			-0.00008** (0.00003)			-0.0287*** (0.0030)
N	8477			8477			8206		

Note: "Method I" is the ex-post regression method with exogenous productivity process (19), "Method II" is the ex-post regression method with endogenous productivity process (20) using the full sample, and "Method III" is the ex-post regression method with endogenous productivity process (20) excluding the transition period. For each method, the first column has no lagged productivity, the second column includes lagged first-order productivity, and the third column includes lagged first-, second- and third-order productivity. All regressions include firm and year-fixed effects. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.